ADITYA UNIVERSITY

ADITYA UNIVERSITY

Energy Resources, Electricity Bill and Safety Measures:
Unit III

Mrs.K.Sandhya Rani

Assistant Professor

Electrical and Electronics Engineering

Aditya University

ADITYA UNIVERSITY

Course Contents

UNIT – III

Energy Resources, Electricity Bill and Safety Measures:

Conventional and non-conventional energy resources; Layout and operation of various Power Generation systems: Hydel, Thermal,Solar and Wind power generation. Calculation of electricity bill for domestic appliances. Working principle of Fuse and Miniature circuit breaker (MCB). Electric Shock, Earthing and its types, Safety Precautions to avoid shock.

ADITYA UNIVERSITY

INTRODUCTION

- Energy is the necessity for the economic development of any country.
- The greater the per capita consumption of energy in a country, the higher is the standard of living of its people.
- The demand for energy resources are increasing day by day in the development of industries, transportation and agricultural activities.
- Power and energy are buzzwords in today's world. Energy exists in different forms in nature but the most important form is the electrical energy.
- The survival of industrial undertakings and our social structures depends primarily on low cost and uninterrupted supply of electrical energy.
- Electricity is emerged as basic necessity with food, shelter and clothing for human being.
- Infact, life without electricity has become highly unimaginable.
- Electric locomotives, heating, cooling, fans, blowers, motors, illumination are some applications that converts electrical energy into useful work.

ENERGY RESOURCES

- Energy resources refer to the various natural sources or technologies used to obtain, produce, and convert energy for various purposes, such as electricity generation, transportation, heating, and industrial processes. Energy resources are classified into two types; such as

(i) conventional energy resources, and

(ii) non-conventional resources.

Conventional Energy Resources

- Conventional energy resources refer to traditional sources of energy that have been widely used for many years to generate electricity and power in various aspects of modern life.
- There resources are typically non-renewable, which means they are limited and will eventually exhaust.
- The most common conventional energy resources include:

(i) Fossil Fuels

- **(a) Coal:** Coal is a black or brownish-black sedimentary rock that is primarily composed of carbon. It has been used for centuries as a source of heat and energy and burned to produce electricity in coal-fired power plants.

Conventional Energy Resources

- **(b) Oil (Petroleum):** Crude oil is refined to produce various products, including gasoline, diesel fuel, and jet fuel. These fuels are used in transportation and for power generation.
- **(c) Natural Gas:** Natural gas is a hydrocarbon gas composed primarily of methane. It is used for heating, power generation, and as a fuel for vehicles and industrial processes.

(ii) Nuclear Energy: Nuclear power is generated by harnessing the heat produced during Nuclear Fission reactions (fission splits a heavy atom into two lighter atoms, while fusion combines two light atoms to form a heavier atom). It is considered a conventional energy source because it has been in use for several decades. Nuclear power plants use uranium or plutonium as fuel.

(iii) Hydropower: Although often considered a renewable energy source, hydropower can also be classified as conventional when it involves large-scale dams and reservoirs. Water is used to turn turbines, generating electricity. This is a mature technology and has been widely used for many years.

Conventional Energy Resources

(iv) Biomass: Biomass energy is derived from organic materials such as wood, crop residues, and animal waste. It can be burned directly for heat or converted into biofuels like ethanol and biodiesel for transportation and power generation.

These conventional energy sources have been the backbone of the global energy industry for many years. However, they have several drawbacks, including environmental concerns (e.g., greenhouse gas emissions, air and water pollution, and habitat disruption) and the fact that they are finite resources. As a result, there has been a growing emphasis on transitioning to cleaner and more sustainable energy sources, such as renewable energy (e.g., solar, wind, and hydro) and improving energy efficiency to reduce our reliance on conventional energy resources and mitigate their negative impacts on the environment.

Non-conventional Energy Resources

Non-conventional energy resources, also known as alternative or renewable energy resources, because they are typically sustainable, environmentally friendly, and have an unlimited supply.

These resources are obtained by using natural processes, which can be replenished in a short amount of time. Here are some examples of non-conventional energy resources:

(i) Solar Energy:

Solar power is generated by capturing energy from the sun using photovoltaic cells or solar panels. It is a clean and virtually unlimited source of energy that can be used for electricity generation and heating. Solar panels can be installed on rooftops, in solar farms, and even integrated into building materials.

Non-conventional Energy Resources

(ii) Wind Energy: Wind turbines convert the kinetic energy of moving air into electricity. Wind power is considered a renewable resource because wind is a natural occurrence that will continue as long as the sun shines. Wind farms are often located in areas with consistent wind patterns.

(iii) Hydropower (Small-Scale): In addition to conventional large-scale hydropower, there are smaller-scale hydropower systems that can be installed in rivers of streams to generate electricity. These micro-hydro or mini-hydro systems are less disruptive to ecosystems than large dams.

(iv) Geothermal Energy: Geothermal power plants use heat from within the Earth to generate electricity. This heat can be sourced from hot water reservoirs, geysers, or even hot rocks deep underground. Geothermal energy is reliable and has a low environmental impact.

Non-conventional Energy Resources

(v) Biomass Energy (Sustainable Sources): While conventional biomass can have negative environmental impacts, non-conventional biomass energy sources come from sustainably managed forests and agricultural residues. These materials can be used to generate heat, electricity, or biofuels.

(vi) Ocean Energy: Ocean energy includes tidal energy, wave energy, and ocean thermal energy. These technologies harness the power of the ocean's movements and temperature differences to generate electricity. They are still in the early stages of development but have great potential.

(vii) Hydrogen Energy: Hydrogen can be produced through various methods, including electrolysis of water using renewable electricity. It can be used as a clean fuel for vehicles and power generation when produced using non-polluting energy sources.

Non-conventional Energy Resources

(viii) Biofuels: Non-conventional biofuels are produced from organic materials like algae or non-food crops. They can be used to replace traditional fossil fuels in transportation and are considered more environmentally friendly.

(ix) Waste-to-Energy: This involves converting municipal solid waste or organic waste into energy through processes like incineration or anaerobic digestion. It not only reduces waste but also produces energy.

Non-conventional energy resources are seen as more sustainable and environmentally responsible alternatives to conventional energy sources like coal, oil, and natural gas. They play a crucial role in reducing greenhouse gas emissions, mitigating climate change, and transitioning to a cleaner and more sustainable energy future.

HYDROELECTRIC POWER PLANT

- A generating station which utilises the potential energy of water at a high level for the generation of electrical energy is known as **a hydro-electric power station**.
- Hydro-electric power stations are generally located in hilly areas where dams can be built conveniently and large water reservoirs can be obtained. In a hydro-electric power station, water head is created by constructing a dam across a river or lake. From the dam, water is led to a water turbine. The water turbine captures the energy in the falling water and changes the hydraulic energy (i.e., product of head and flow of water) into **mechanical energy at the turbine shaft**. The turbine drives the alternator which converts **mechanical energy into electrical energy**. Hydro-electric power stations are becoming very popular because the reserves of fuels (i.e., coal and oil) are depleting day by day. They have the added importance for flood control, storage of water for irrigation and water for drinking purposes.

Selection of site for hydroelectric power plant

The following points should be taken into account while selecting the site for a hydroelectric power station:

(i) Availability of water: The primary requirement of a hydroelectric power plant is the availability of huge quantity of water, so these plants should be built at a place, where adequate water is available at a good head.

(ii) Storage of water: The water supply from a river or canal varies during the year. This requires storage of water by constructing a dam in order to ensure the power generation throughout the year. The storage helps in equalizing the flow of water so that any excess quantity of water at a certain period of the year can be made available during times of very low flow in the river.

Selection of site for hydroelectric power plant

(iii) Availability of land: The land available should be cheap in cost and rocky in order to withstand the weight of the large building and heavy machinery.

(iv) Water pollution: Polluted water may cause excessive corrosive and damage to the metallic structures. Hence, availability of good quality of water is essential.

(v) Transportation facility: The site selected for a hydroelectric plant should be accessible by rail and road so that necessary equipment and machinery can be easily transported.

ADITYA UNIVERSITY

Layout diagram of hydroelectric power plant

The main components of a hydroelectric power plant as shown in figure are as listed below:

(i) Dam

(ii) Reservoir

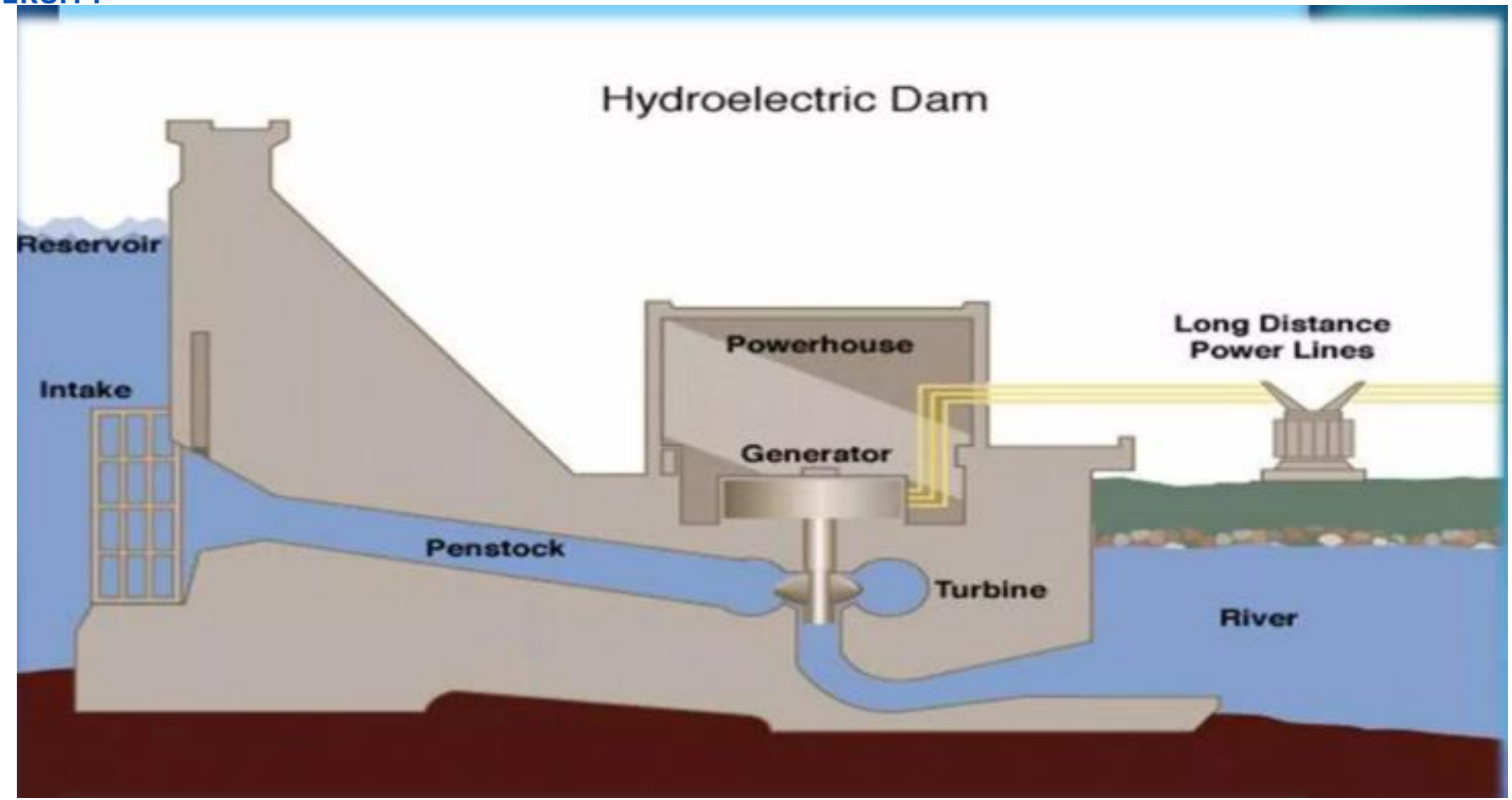
(iii) Penstock

(iv) Surge Tank

(v) Power House

(a) Hydraulic turbine (Prime mover)

(b) Generators and Step-up Transformers


Hydroelectric Dam
Reservoir
Intake
Powerhouse
Generator
Penstock
Turbine
Long Distance
Power Lines
River

Main components of a hydroelectric power plant

i) **Dam:** A dam is a barrier which stores water and creates water head. The purpose of the dam is to store the water and to regulate the outgoing flow of water. The also helps to increase the head water.

ii) **Reservoir:** It is a basic requirement of hydro electric power plant the purpose is to store water which may be used to run the prime mover to produce electrical power. It stores water during rainy season and supplies the water during dry season.

iii) **Penstock:** Penstock is a closed pipe of steel or concrete for supplying water under pressure to the turbine. There are two types of penstock, low pressure and high pressure. The low pressure type consists of canal, a flume or pipe. The high pressure type consists of steel pipes, which can take water under pressure.

Main components of a hydroelectric power plant

(iv) Surge tank: A surge tank is a small reservoir or tank in which water level rises or falls to reduce the pressure swings in the penstock. A surge tank is located near the beginning of the penstock. It reduces the distance between the free water surface in the dam and the turbine, thereby reducing the water-hammer effect on penstock.

Water-hammer effect: The water hammer is defined as the change in pressure rapidly above or below normal pressure caused by sudden change in the rate of water flow through the pipe, according to the demand of prime mover i.e. turbine.

(v) Power house: It is generally located at the foot of the dam and near the storage reservoir. If the power house is near the dam, the loss of head due to friction in the penstock would be less. A power house consists of two main parts, hydraulic and electric equipment's.

Main components of a hydroelectric power plant

(a) **Hydraulic turbine (Prime mover):** The hydraulic turbine converts the kinetic energy of water into mechanical energy. The mechanical energy (rotation) available on the turbine shaft is coupled to the shaft of an electric generator. The water after performing the work on turbine blades is discharged through the draft tube. The prime movers which are in common use are **Pelton wheel, Francis turbine and Kaplan turbine**.

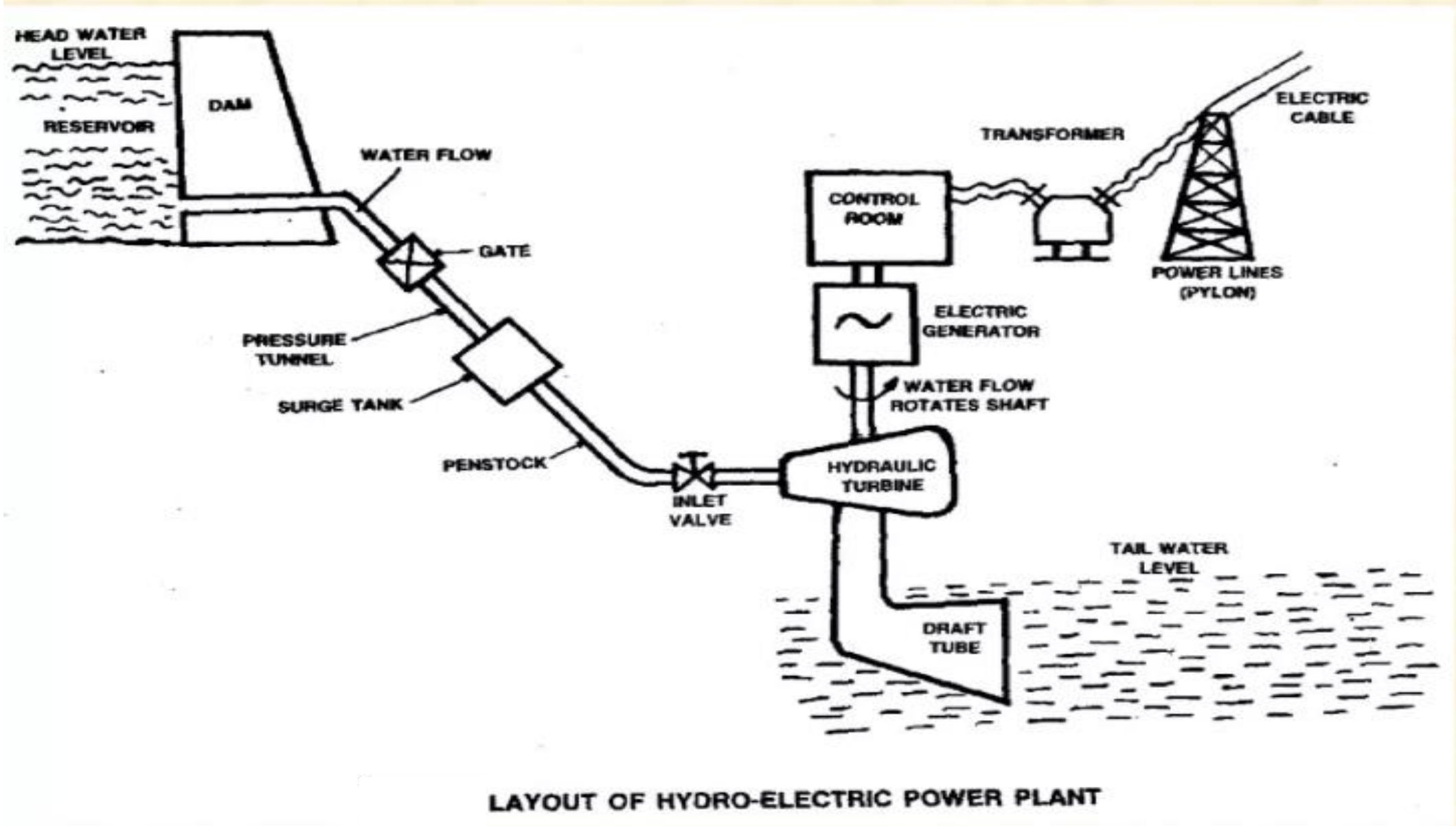
Draft tube: It is connected to the outlet of the turbine. It allows the turbine to be placed above the tail water level. Tail water level is the water level after the discharge from the turbine. The discharged water is sent to the river, thus the level of the river is the tail water level.

Main components of a hydroelectric power plant

(b) Generators and step-up transformers: As the water rushes through the turbine, it spins the turbine shaft, which is coupled to the electric generator. The generator has a rotating electromagnet called a rotor and a stationary part called a stator. The rotor creates a magnetic field that produces an electric charge in the stator. The charge is transmitted as electricity. The step-up transformer increases the voltage and current coming from the stator

Operation of a hydroelectric power plant

- Although a hydro-electric power station simply involves the conversion of hydraulic energy into electrical energy, yet it embraces many arrangements for proper working and efficiency.
- The schematic arrangement of a modern hydro-electric plant is shown in Fig.
- The dam is constructed across a river or lake and water from the catchment area collects at the back of the dam to form a reservoir.
- A pressure tunnel is taken off from the reservoir and water brought to the valve house at the start of the penstock.
- The valve house contains main sluice valves and automatic isolating valves.
- The former controls the water flow to the power house and the latter cuts off supply of water when the penstock bursts.



LAYOUT OF HYDRO-ELECTRIC POWER PLANT

Operation of a hydroelectric power plant

- From the valve house, water is taken to water turbine through a huge steel pipe known as penstock.
- The water turbine converts hydraulic energy into mechanical energy.
- The turbine drives the alternator which converts mechanical energy into electrical energy.
- A surge tank (open from top) is built just before the valve house and protects the penstock from bursting in case the turbine gates suddenly close due to electrical load being thrown off.
- When the gates close, there is a sudden stopping of water at the lower end of the penstock and consequently the penstock can burst like a paper log.
- The surge tank absorbs this pressure swing by increase in its level of water.

ADITYA UNIVERSITY

ADVANTAGES

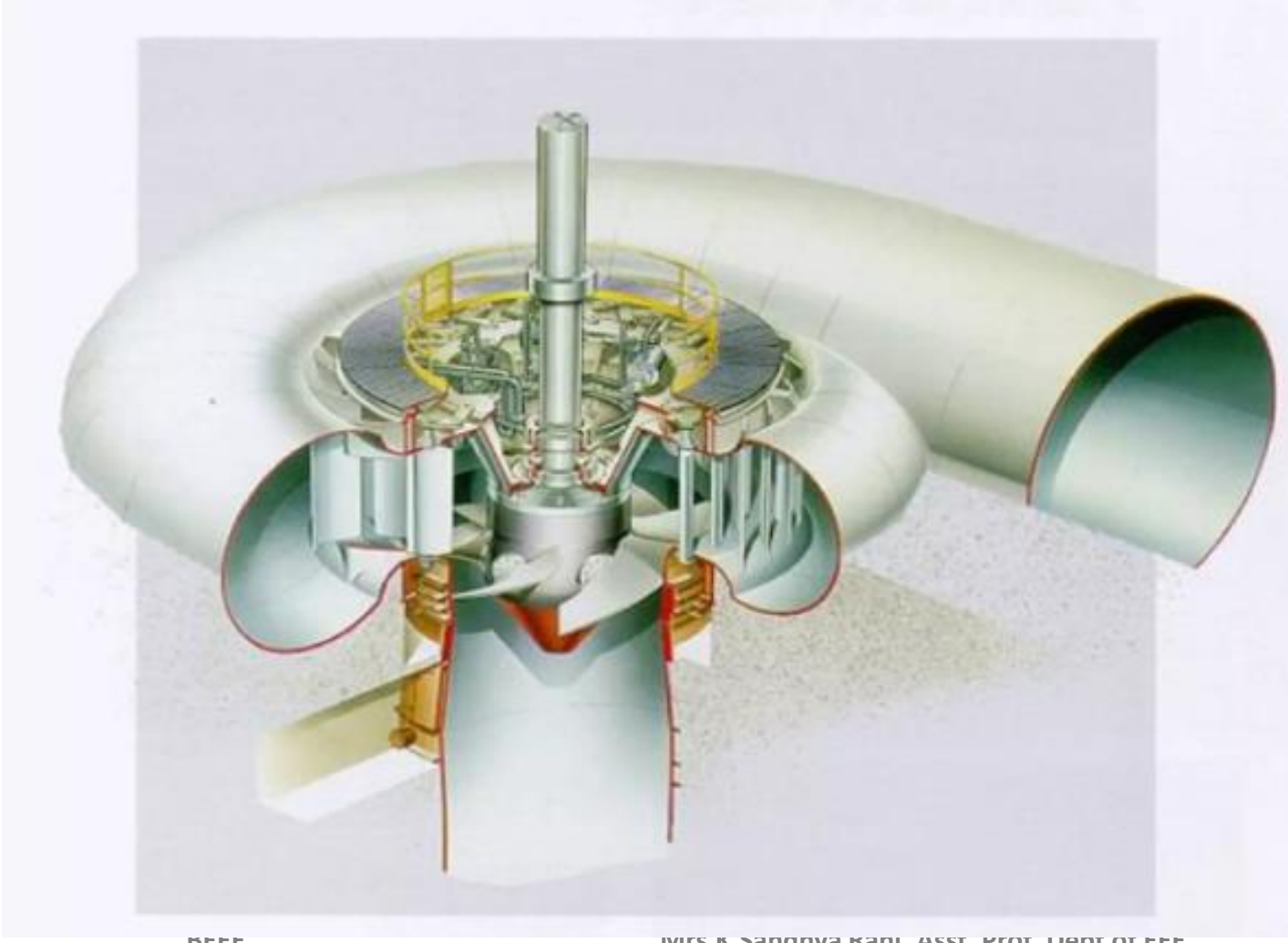
- No fuel charges.
- Less supervising staff is required.
- Maintenance & operation charges are very low.
- Running cost of the plant is low.
- No fuel transportation is required.
- No ash & flue gas problem & does not pollute the atmosphere.
- These plants are used for flood control & irrigation purpose.
- Long life in comparison with the thermal & nuclear power plants

DISADVANTAGES

- It involves high capital cost due to construction of dam.
- There is uncertainty about the availability of huge amount of water due to dependence on weather conditions.
- Skilled and experienced hands are required to build the plant.
- It requires high cost of transmission lines as the plant is located in hilly areas which are quite away from the consumers.

Main components of a hydroelectric power plant

Kaplan turbine



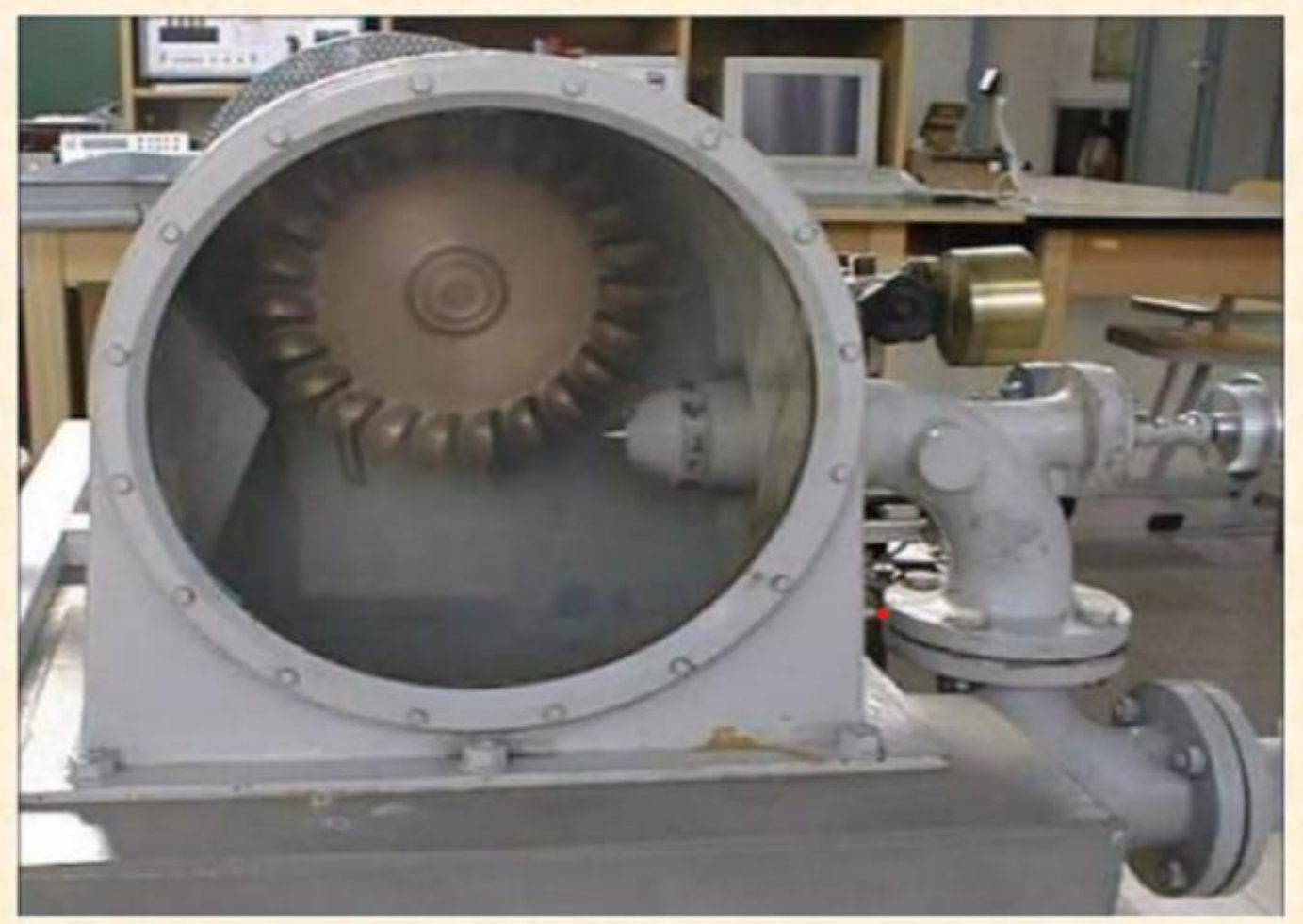
Main components of a hydroelectric power plant

Francis turbine



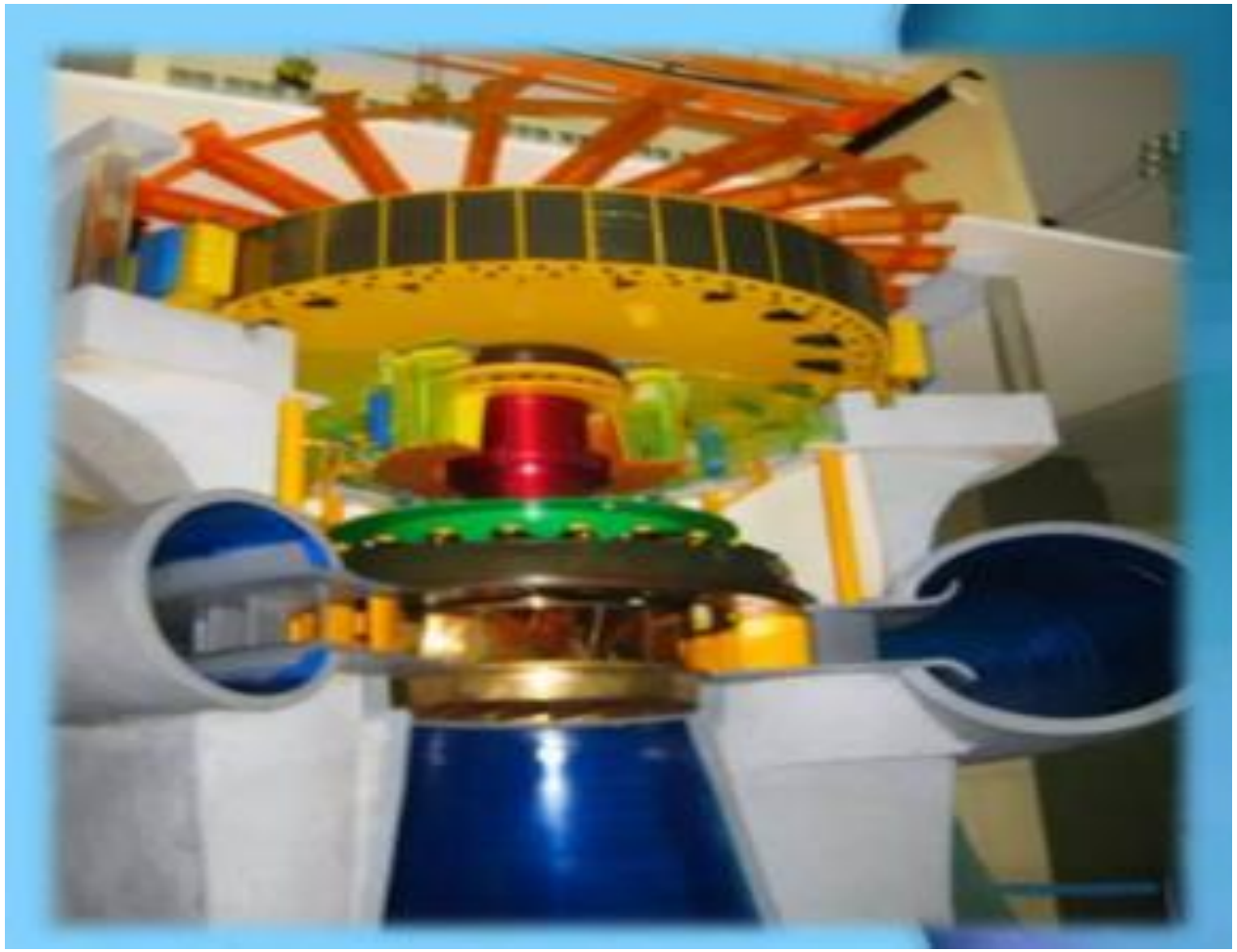
Main components of a hydroelectric power plant

Pelton wheel



Main components of a hydroelectric power plant

Generator



Main components of a hydroelectric power plant

Penstock



Main components of a hydroelectric power plant

Transformer



Steam Power Station (Thermal Station)

- A generating station which converts **heat energy of coal combustion into electrical energy** is known as a steam power station.
- A steam power station basically works on the **Rankine cycle**. Steam is produced in the boiler by utilising the heat of coal combustion.
- The steam is then expanded in the prime mover (i.e., steam turbine) and is condensed in a condenser to be fed into the boiler again.
- The steam turbine drives the alternator which converts **mechanical energy of the turbine into electrical energy**.
- This type of power station is suitable where coal and water are available in abundance and a large amount of electric power is to be generated.

Choice of Site for Steam Power Stations

In order to achieve overall economy, the following points should be considered while selecting a site for a steam power station :

(i) Supply of fuel: The steam power station should be located near the coal mines so that transportation cost of fuel is minimum. However, if such a plant is to be installed at a place where coal is not available, then care should be taken that adequate facilities exist for the transportation of coal.

(ii) Availability of water: As huge amount of water is required for the condenser, therefore, such a plant should be located at the bank of a river or near a canal to ensure the continuous supply of water.

(iii)Transportation facilities: A modern steam power station often requires the transportation of material and machinery. Therefore, adequate transportation facilities must exist i.e., the plant should be well connected to other parts of the country by rail, road. etc.

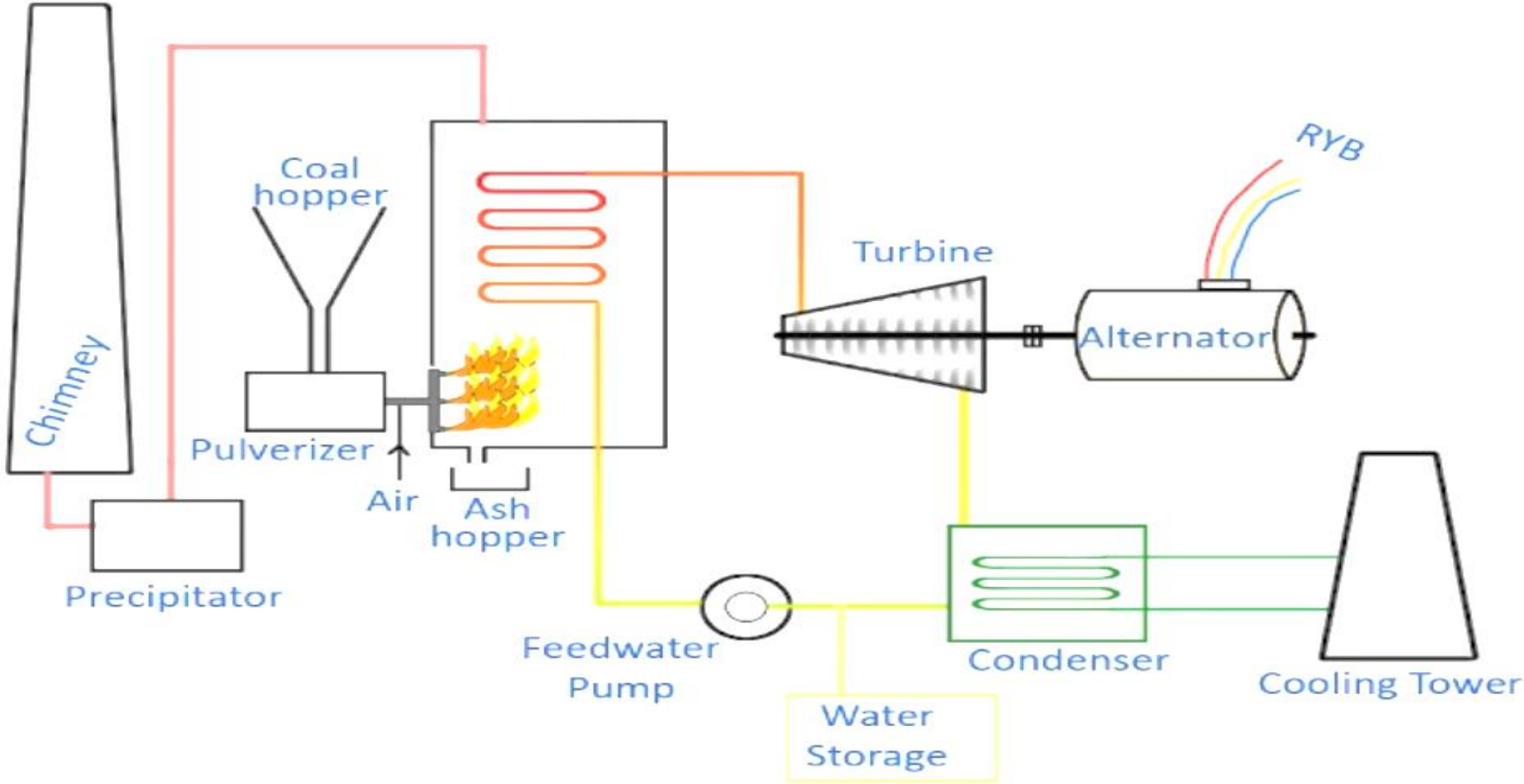
Choice of Site for Steam Power Stations

(iv) Cost and type of land: The steam power station should be located at a place where land is cheap and further extension, if necessary, is possible. Moreover, the bearing capacity of the ground should be adequate so that heavy equipment could be installed.

(v) Nearness to load centres: In order to reduce the transmission cost, the plant should be located near the centre of the load. This is particularly important if d.c. supply system is adopted. However, if a.c. supply system is adopted, this factor becomes relatively less important. It is because a.c. power can be transmitted at high voltages with consequent reduced transmission cost. Therefore, it is possible to install the plant away from the load centres, provided other conditions are favorable.

(vi) Distance from populated area: As huge amount of coal is burnt in a steam power station, therefore, smoke and fumes pollute the surrounding area. This necessitates that the plant should be located at a considerable distance from the populated areas.

Layout of a Thermal Power Plant



Layout of a Thermal Power Plant

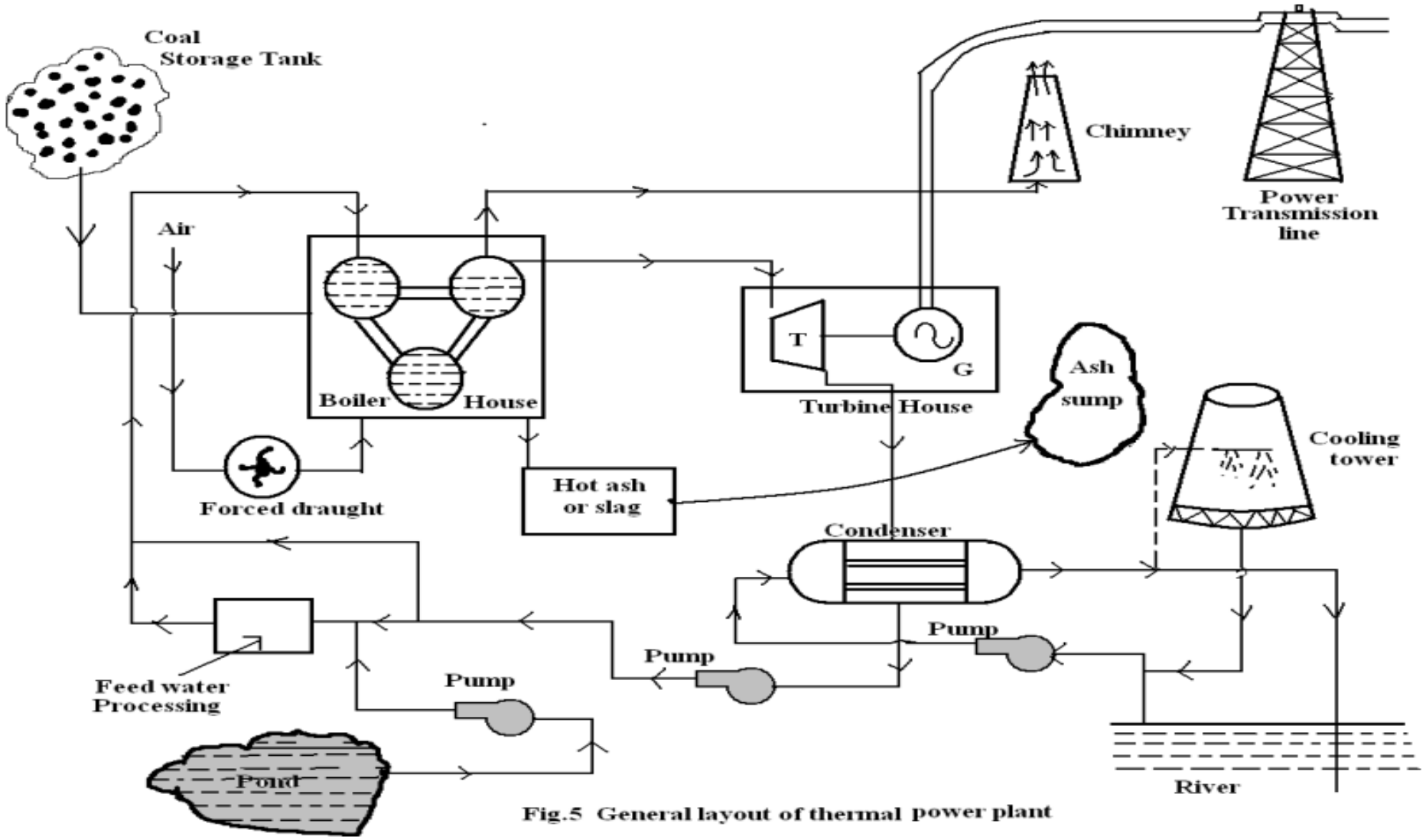


Fig.5 General layout of thermal power plant

Major Components of Steam Power Stations

Coal: In a coal based thermal power plant, coal is transported from coal mines to the generating station. Generally, bituminous coal or brown coal is used as fuel. The coal is stored in either **'dead storage' or in 'live storage'.** Dead storage is generally 40 days backup coal storage which is used when coal supply is unavailable. Live storage is a raw coal bunker in boiler house. The coal is cleaned in a magnetic cleaner to filter out if any iron particles are present which may cause wear and tear in the equipment. The coal from live storage is first crushed in small particles and then taken into pulverize to make it in powdered form. Fine powdered coal undergoes complete combustion, and thus pulverized coal improves efficiency of the boiler. The ash produced after the combustion of coal is taken out of the boiler furnace and then properly disposed. Periodic removal of ash from the boiler furnace is necessary for the proper combustion.

Major Components of Steam Power Stations

Boiler: The mixture of pulverized coal and air (usually preheated air) is taken into boiler and then burnt in the combustion zone. On ignition of fuel a large fireball is formed at the center of the boiler and large amount of heat energy is radiated from it. The heat energy is utilized to convert the **water into steam at high temperature and pressure**. Steel tubes run along the boiler walls in which **water is converted in steam**. The flue gases from the boiler make their way through superheater, economizer, air preheater and finally get exhausted to the atmosphere from the chimney.

Major Components of Steam Power Stations

- **Superheater:** The superheater tubes are hanged at the hottest part of the boiler. The saturated steam produced in the boiler tubes is superheated to about 540 °C in the superheater. The superheated high pressure steam is then fed to the steam turbine.

- **Economizer:** An economizer is essentially a feed water heater which heats the water before supplying to the boiler.

- **Air pre-heater:** The primary air fan takes air from the atmosphere and it is then warmed in the air pre-heater. Pre-heated air is injected with coal in the boiler. The advantage of pre-heating the air is that it improves the coal combustion.

Major Components of Steam Power Stations

Steam turbine: High pressure super heated steam is fed to the steam turbine which causes turbine blades to rotate. Energy in the **steam is converted into mechanical energy** in the steam turbine which acts as the prime mover. The pressure and temperature of the steam falls to a lower value and it expands in volume as it passes through the turbine. The expanded low pressure steam is exhausted in the condenser.

Condenser: The exhausted steam is condensed in the condenser by means of cold water circulation. Here, the steam loses it's pressure as well as temperature and it is **converted back into water**. Condensing is essential because, compressing a fluid which is in gaseous state requires a huge amount of energy with respect to the energy required in compressing liquid. Thus, condensing increases efficiency of the cycle.

Major Components of Steam Power Stations

Alternator: The steam turbine is coupled to an alternator. When the turbine rotates the alternator, **electrical energy is generated**. This generated electrical voltage is then stepped up with the help of a transformer and then transmitted where it is to be utilized.

Feed water pump: The condensed water is again fed to the boiler by a feed water pump. Some water may be lost during the cycle, which is suitably supplied from an external water source.

Working of Steam Power Stations

Steam is generated in the boiler of thermal power plant using the heat of the fuel burned in the combustion chamber. The steam generated is passed through steam turbine where part of its thermal energy is converted into mechanical energy which is further used for generating electric power.

Advantages of Steam Power Stations

- It needs low initial investment and less time to commission the plant.
- The cost of the Steam power plant is lower than several power plants.

Disadvantages of Steam Power Stations

- The life and effectiveness of the steam power plant are more concise when compared to Hydel power plant.
- Transport of fuel is a major problem.
- The cost of power generation is higher than hydropower.
- Air pollution is a major difficulty.
- Coal may be depleted by gradual use.

SOLAR POWER GENERATION

- Solar power generation is the process of **converting sunlight into electricity or heat** using photovoltaic (PV) panels or solar thermal systems.
- Solar power is a renewable and sustainable energy source that has gained popularity due to its environmental benefits and decreasable cost.
- The layouts and operation of a solar power generation system can vary depending on its scale and purpose, but we will provide a general overview of the components and how they work together in a typical grid-tied solar power system.

Selection of Site for Solar Power Generation

Selecting the right site for solar power generation is critical to maximize the efficiency and effectiveness of solar energy system. Several factors should be considered when choosing a site for a solar power installation, whether it's a residential, considered, or utility-scale project. The various factors are as listed below:

(i) Solar resource potential: The amount of sunlight a site receives throughout the year is a primary consideration. A location with a lot of direct sunlight and a high solar insolation is needed.

(ii) Orientation and tilt: The orientation and tilt angle of the solar panels significantly impact energy production. Solar panels should ideally face south for maximum sun exposure. The tilt angle should be optimized based on the site's latitude.

Selection of Site for Solar Power Generation

(iii) Shading: Avoid shading from nearby structures, trees, or other obstructions. Even small amounts of shading can reduce the efficiency of solar panels.

(iv) Available space: Assess the available space for solar panel installation. Rooftop systems utilize existing structures, while ground-mounted systems require open land.

(v) Grid access: If you plan to connect your solar power system to the electrical grid, ensure that there is a reliable grid connection available. Proximity to the grid connection point can affect installation costs.

(vi) Land use and zoning: Check local land use regulations and zoning codes to ensure that solar installations are allowed in the chosen area. Some areas may have restrictions on land use or may require permits for solar projects.

Selection of Site for Solar Power Generation

(vii) Environmental impact: Evaluate the potential environmental impact of solar installation. Ensure that it doesn't harm protected species, disrupt ecosystems, or violate environmental regulations. Environmental assessments and permits may be necessary.

(viii) Topography and soil conditions: The site's topography and soil conditions can affect the cost and feasibility of solar installation. Steep slopes, rocky terrain, or poor soil quality may require additional engineering and construction work.

(ix) Weather conditions: Consider local weather patterns and climate conditions. Extreme weather events, such as hurricanes or heavy snowfall, may impact the design and durability of the solar power system.

(x) Accessibility: Ensure that the site is easily accessible for equipment delivery, installation, and maintenance. Accessibility can affect installation costs and ongoing upkeep.

Selection of Site for Solar Power Generation

(xi) Local incentives and regulations: Research local incentives, tax credits, and regulatory policies that may impact the financial feasibility of solar project. Some regions offer incentives to encourage solar adoption.

(xii) Community and stakeholder support: Engage with the local community and stakeholders to address concerns and build support for the solar project. Public opinion and community support can influence the permitting process and project success.

(xii) Long-term goals: Consider the long-term energy goals. Determine if the selected site can accommodate future expansions or additions to your solar power system.

Grid-tied Solar Power System

Grid-tied solar system generally consists of the following components as shown in figure

(i) Solar Panels (Photovoltaic Cells)

(ii) Mounting Structure

(iii) Inverters

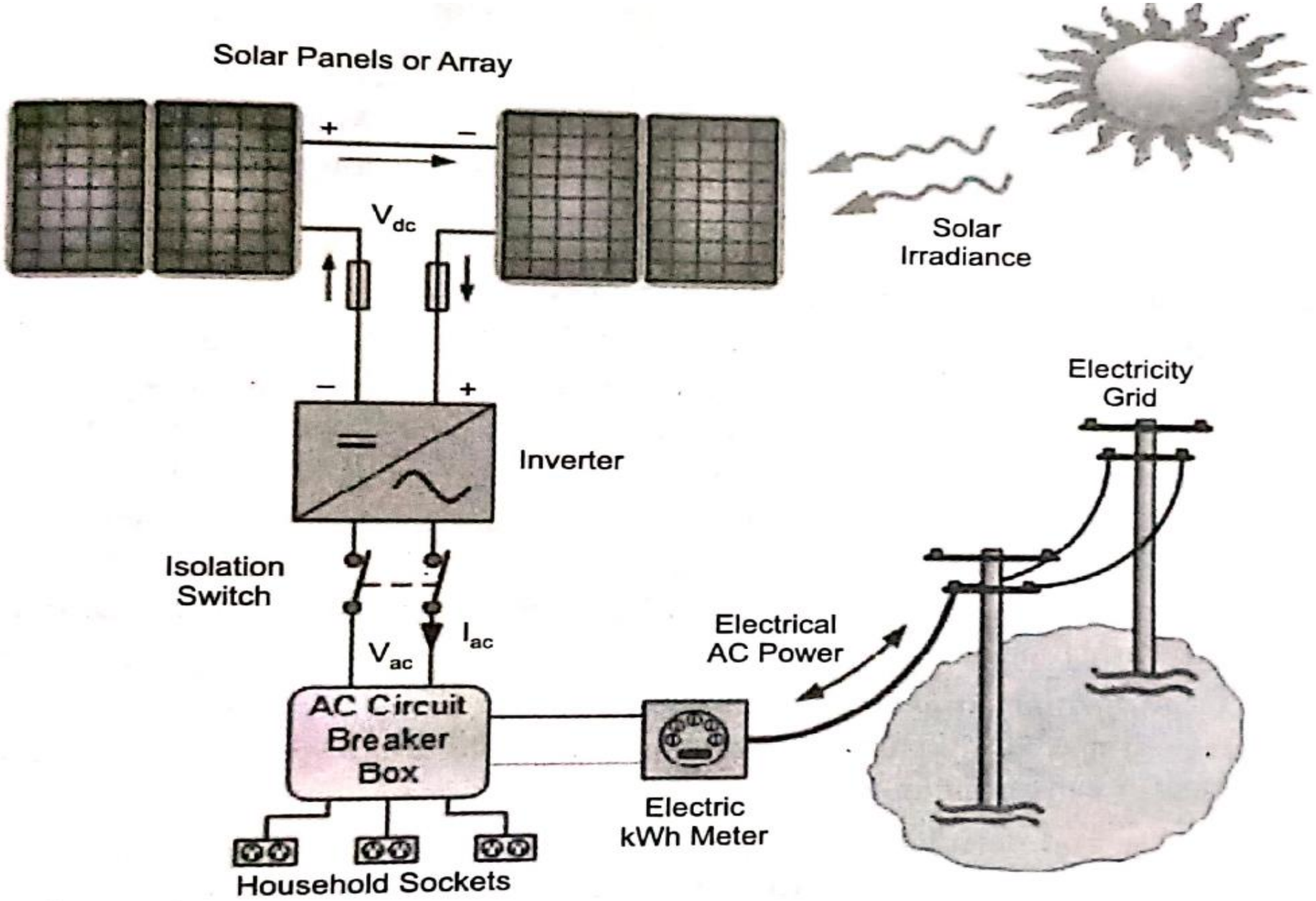
(iv) Electrical Panel

(v) Net Meter

(vi) Utility Grid Connection

(vii) Monitoring System

Grid-tied Solar Power System



Grid-tied Solar Power System

(i) **Solar Panels (Photovoltaic Cells):** Solar panels are the heart of a solar power system. They consist of photovoltaic cells made of semiconductor materials, typically silicon. When exposed to sunlight, these cells generate direct current (DC) electricity. Solar panels can be mounted on various surfaces, such as rooftops, ground mounts, or tracking systems to maximize sun exposure.

(ii) **Mounting Structure:** Solar panels are typically mounted on a structure that positions them at an optimal angle and orientation to the sun. Rooftop systems are attached to the roof, while ground-mounted systems are installed on the ground. Tracking systems follow the sun's path throughout the day to maximize energy production.

(iii) **Inverters:** The electricity generated by solar panels is in DC form, but most household appliances and the electrical grid operate on alternating current (AC). Inverters are used to convert the DC electricity produced by the solar panels into AC electricity.

Grid-tied Solar Power System

(iv) Electrical Panel: The AC electricity produced by the inverter is then sent to the electrical panel or breaker box of the building. It can be used to power appliances and electrical loads within the building.

(v) Net Meter: If the solar power system is grid-tied (connected to the electrical grid), a net meter is installed. This bi-directional meter measures the electricity flowing both from the grid to the building and from the solar system to the grid. When the solar system generates more electricity than is needed, the excess is sent back to the grid, and the meter records this surplus.

(vi) Utility Grid Connection: A grid-tied solar power system remains connected to the local utility grid. During periods of high solar production, excess electricity is fed into the grid. Conversely, when solar production is low (e.g., at night), electricity is drawn from the grid.

(vii) Monitoring System: Many solar power systems come with monitoring equipment that allows homeowners or system operators to track the system's performance, energy production, and overall health. This helps identify issues and optimize system efficiency.

Operation of Grid-tied solar power system

Sunlight Capture: Solar panels capture sunlight and convert it into electricity through the photovoltaic effect. The more direct and intense the sunlight, the more electricity the parrels generate.

Inverter Conversion: The DC electricity produced by the solar panels is sent to the inverter, where it is converted into AC electricity suitable for household use.

Power Consumption: The AC electricity produced by the inverter is used to power the building's electrical loads. If the solar system generates more electricity than the building consumes, the excess is exported to the grid.

Grid Interaction: In a grid-tied system, any excess electricity is sent to the utility grid, and the homeowner may receive credits or compensation for this surplus energy.

Night and Low Sunlight: During nighttime and periods of low sunlight, electricity is drawn from the grid to meet the building's power needs.

Operation of Grid-tied solar power system

Monitoring and Maintenance: The system's performance is monitored to ensure it operates efficiently. Routine maintenance may include cleaning panels, checking connections, and addressing any issues.

Solar power generation systems offer renewable and clean energy, reduce electricity bills, and can have a positive environmental impact by reducing greenhouse gas emissions .The operation of off-grid systems is similar but includes battery storage to provide electricity during periods of low or no sunlight.

Advantage of Grid-tied solar power system

- Solar power plants use renewable and clean energy that does not emit greenhouse gases or pollutants.
- Solar power plants can reduce dependence on fossil fuels and enhance energy security and diversity.
- Solar power plants can provide electricity in remote areas where grid connection is not feasible or reliable.
- Solar power plants can create local jobs and economic benefits for communities and regions.
- Solar power plants can benefit from various incentives and policies that support renewable energy development and deployment.

Disadvantage of Grid-tied solar power system

- Solar power plants require large land areas and may have environmental impacts on wildlife, vegetation, and water resources.
- Solar power plants have high initial capital costs and long payback periods compared to conventional power plants.
- Solar power plants have low capacity factors and depend on weather conditions and diurnal cycles that affect their output and reliability.
- Solar power plants need backup or storage systems to ensure a continuous supply of electricity during periods of low or no sunlight.
- Solar power plants face technical challenges such as grid integration, interconnection, transmission, and distribution.

WIND POWER GENERATION

Wind power generation, often referred to as wind energy or wind electricity, is the process of converting the kinetic energy of wind into electrical power. Wind turbines are the primary technology used for harnessing wind energy, and they have become an increasingly important source of renewable energy worldwide.

Selection of site for wind power generation

Site selection for wind power generation requires a thorough assessment of technical, environmental, regulatory, and social factors. It's often advisable to work with wind energy professionals and conduct a comprehensive wind resource assessment to ensure that the chosen site is suitable for your specific wind energy project. The various factors are as listed below:

(i) **Wind resource assessment:** The primary factor to assess is the wind resource at the site. We will need to gather data on wind speeds, directions, and turbulence at various heights above the ground. This data is typically collected over at least a year, using anemometers and wind vanes.

(ii) **Wind turbine siting:** The wind turbine should be located where it can capture the highest and most consistent wind speeds. Avoid areas with significant turbulence, such as behind buildings or near large obstructions.

Selection of site for wind power generation

(iii) Zoning and permits: Check local zoning regulations and permitting requirements. Ensure that the site is zoned for wind energy projects and obtain the necessary permits before proceeding.

(iv) Environmental impact: Assess the potential environmental impact of the wind energy project. Consider the effects on local wildlife, particularly birds and bats, as well as the impact on natural landscapes and ecosystems. Environmental assessments and permits may be required.

(v) Access and infrastructure: Ensure that the site is accessible for the delivery, installation, and maintenance of wind turbines and related infrastructure. Adequate road access, crane access, and electrical connections to the grid are essential.

Selection of site for wind power generation

(vi) Grid connection: Confirm the feasibility of connecting the wind energy system to the electrical grid. Analyze the capacity of the local grid to accommodate the energy produced and the requirements for grid interconnection.

(vii) Meteorological data: In addition to wind resource data, consider other meteorological factors such as temperature, humidity, and air density, as these can affect turbine performance and efficiency.

(viii) Turbine height and size: Choose an appropriate turbine height and size, based on the wind resource and site conditions. Taller turbines can access higher wind speeds, but they may have stricter permitting requirements and higher costs.

(ix) Aviation and airspace considerations: Ensure that the site complies with aviation and airspace regulations. Wind turbines can interfere with air traffic, so you may need to coordinate with aviation authorities.

Selection of site for wind power generation

(x) Community and stakeholder engagement: Engage with the local community and stakeholders to address concerns, build support, and foster positive relationships Public opinion and local support can impact the permitting process and project

(xi) Long-term viability: Consider the long-term viability of the site. Assess how the wind resource may change over time, as well as the expected lifespan of the wind turbines. Plan for regular maintenance and potential repowering.

Layout of wind power generation

Wind power generation system generally consists of the following components as shown in figure :

(i) Wind Turbines

(ii) Energy Conversion System

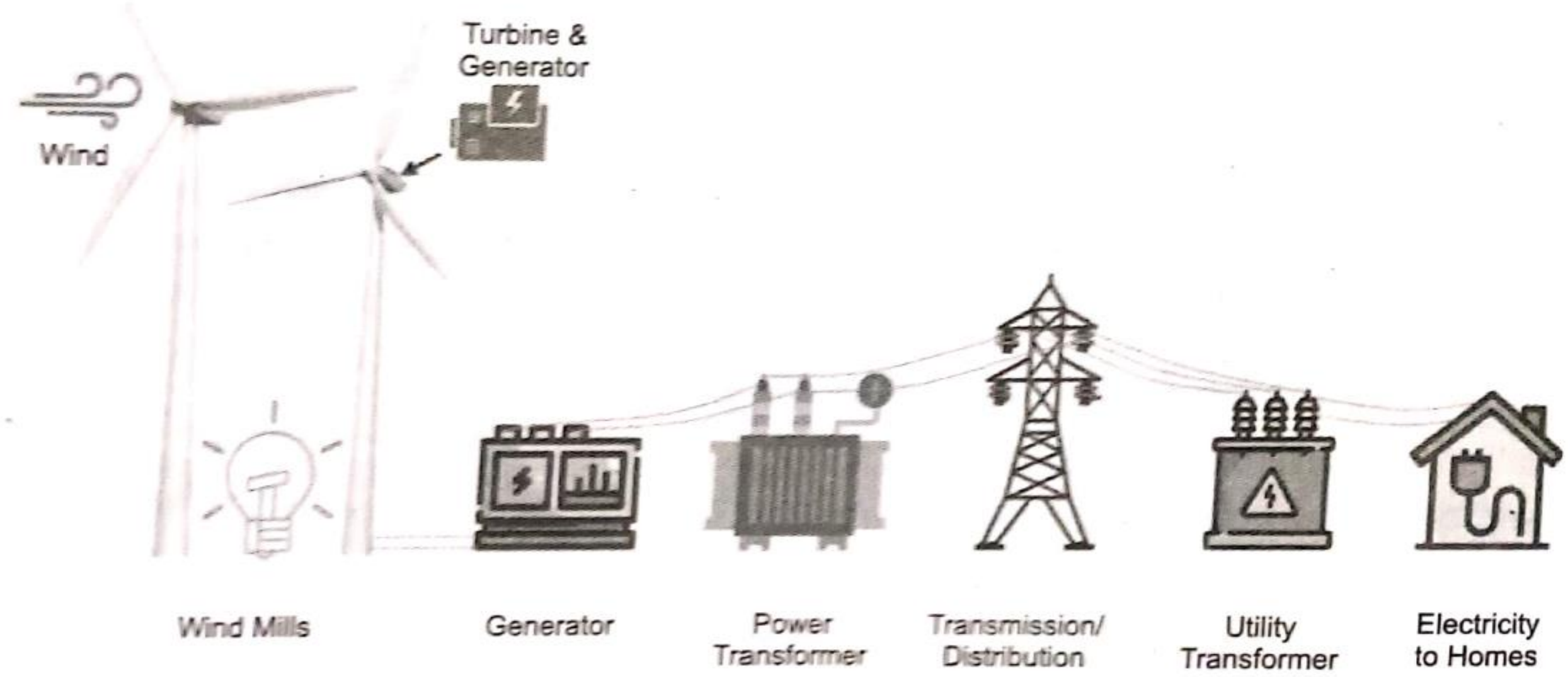
(iii) Electrical System

(iv) Grid Connection

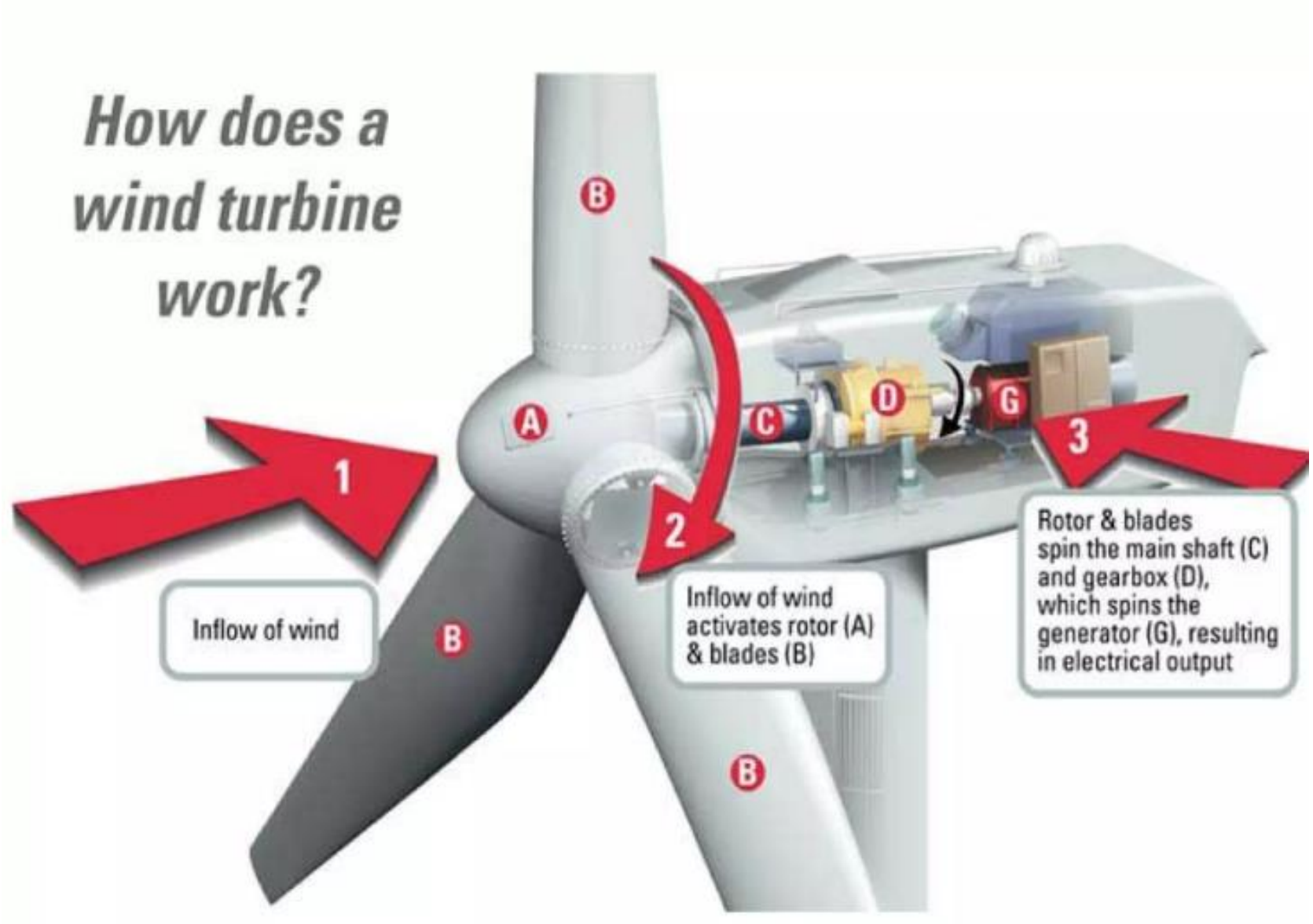
(v) Energy Storage (Optional)

(vi) Monitoring and Control unit

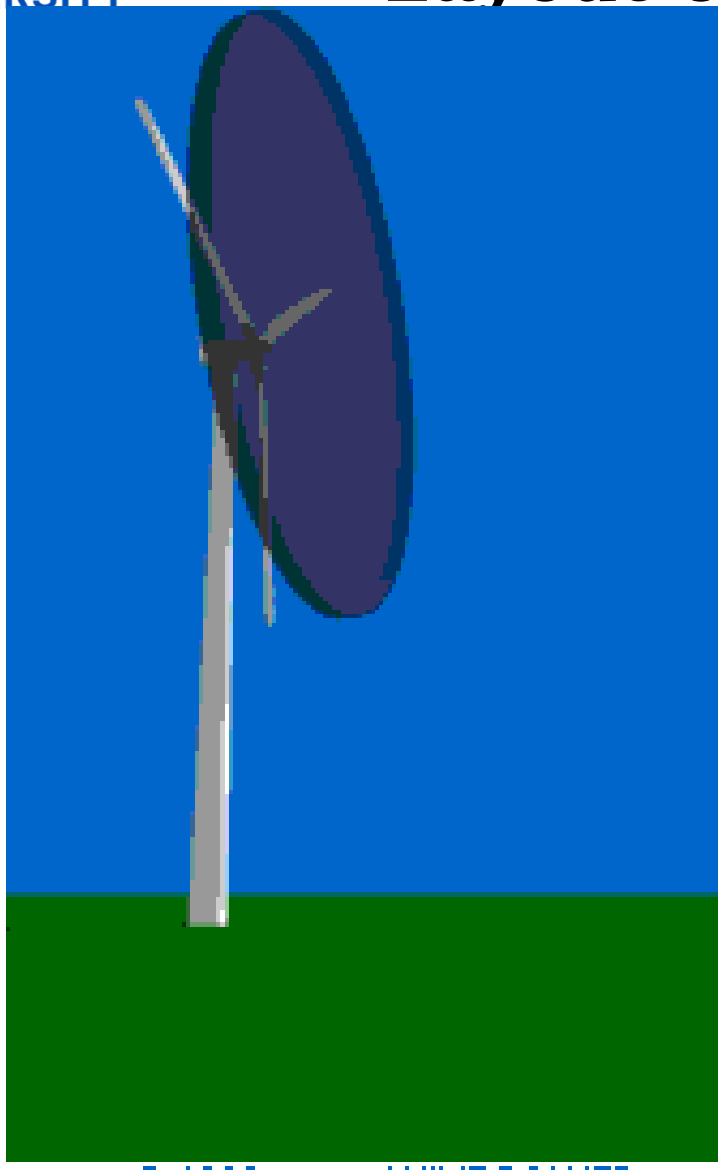
Layout of wind power generation



Layout of wind power generation



Layout of wind power generation





Layout of wind power generation

(i) **Wind turbines:** Wind turbines are the primary components of a wind power generation system. They consist of large rotor blades mounted on a hub, which is connected to a generator. The kinetic energy of the wind causes the blades to rotate, which in turn drives the generator. Wind turbines are strategically placed in locations with a consistent and strong wind resource. Factors such as wind speed, direction, and turbulence are considered when determining optimal placement within a wind farm.

(ii) **Energy conversion system:** As the wind flows over the rotating blades, it causes them to turn, converting kinetic energy into mechanical energy. The mechanical rotation is used to turn a generator, typically an alternator, which converts mechanical energy into electrical energy.

Layout of wind power generation

(iii) Electrical system: The electricity generated by the wind turbine's generator is initially in the form of alternating current (AC). In many cases, wind turbines are connected to an electrical system that includes transformers, switchgear, and inverters to adjust voltage levels and convert AC power into a suitable form for transmission and distribution.

(iv) Grid connection: Wind power systems are often connected to the electrical grid, allowing the generated electricity to be distributed to homes, businesses, and industries. Grid-connected wind farms use transmission lines to transport electricity to consumers.

(v) Energy storage: Some wind power installations incorporate energy storage systems, such as batteries, to store excess electricity during periods of high wind. Energy storage helps ensure a consistent and reliable energy supply, even when the wind isn't blowing.

(vi) Monitoring and control unit: Wind power systems are equipped with monitoring control system that continuously track wind conditions, turbines performance. These systems optimize turbine operation and can diagnose issues for maintenance.,

Operation of Wind Power Generation System

Wind Capture: As the wind blows, it flows over the blades of the wind turbine, causing them to rotate. The shape and design of the blades are engineered to efficiently capture the kinetic energy of the wind.

Mechanical Rotation: The mechanical rotation of the turbine's blades is transferred to the generator's rotor. This rotor is connected to a shaft, which spins inside the generator.

Energy Conversion: Within the generator, the mechanical energy is converted into electrical energy through electromagnetic induction. This process produces alternating current (AC) electricity.

Grid Interaction: In a grid-connected system, the AC electricity is synchronized with the grid's frequency and voltage and is fed into the electrical grid. This electricity is then distributed to consumers.

Energy Storage (If Applicable): If an energy storage system is present, excess electricity generated during periods of high wind can be stored for later use.

Operation of Wind Power Generation System

Monitoring and Control: The wind power system continuously monitors wind conditions, turbine performance, and electrical output. Automated control systems adjust the turbine's orientation (yaw), blade pitch, and other parameters to optimize energy capture and ensure safe operation.

Wind power generation offers numerous advantages, including sustainability, reduced greenhouse gas emissions, and renewable energy production. However, challenges such as intermittency, land use, and visual impact should be considered when planning and operating wind power projects. Wind power is becoming a more viable source of clean energy because of improvements in wind turbine technology, energy storage, and grid integration.

Advantages of Wind Power Generation System

- Environmental
- Economic Development
- Fuel diversity & Conservation
- Cost stability

Disadvantages of Wind Power Generation System

- Birds – A serious obstacle
- Noise disturbance
- Cost of wind turbine
- Threat to wild life
- Wind can be never predicted
- Visual Impact
- Suited to particular areas or regions

Power Rating of Household Appliances

- In our daily lives, household appliances play a vital role in enhancing convenience, comfort, and efficiency. However, each appliance consumes a specific amount of electrical energy, which is quantified through power ratings. Power ratings, measured in watts (W) or kilowatts (kW), indicate the amount of electrical power an appliance requires to operate effectively.
- Understanding power ratings is essential for several reasons. Firstly, they help consumers estimate the energy consumption of their devices, allowing for better budgeting of electricity costs.
- Secondly, knowledge of power ratings can guide decisions when purchasing new appliances, as energy-efficient models can significantly reduce long-term expenses and environmental impact.
- Additionally, awareness of power ratings aids in managing overall electricity consumption within a household. By knowing which appliances use more power, homeowners can optimize usage patterns, particularly during peak tariff hours, thereby lowering their electricity bills.

Power Rating of Household Appliances

- Heating and Cooling:
  - ✓ Air Conditioner: 1,000 - 2,500 watts
  - ✓ Electric Heater: 1,000 - 3,000 watts
  - ✓ Ceiling Fan: 50 - 100 watts
- Kitchen Appliances:
  - ✓ Refrigerator: 100 - 800 watts
  - ✓ Microwave Oven: 600 - 1,200 watts
  - ✓ Electric Oven: 2,000 - 5,000 watts
  - ✓ Dishwasher: 1,200 - 1,500 watts
  - ✓ Toaster: 800 - 1,500 watts
  - ✓ Blender: 300 - 1,000 watts
  - ✓ Slow Cooker: 200 - 500 watts
  - ✓ Coffee Maker: 600 - 1,200 watts

Power Rating of Household Appliances

- Heating and Cooling:
  - ✓ Air Conditioner: 1,000 - 2,500 watts
  - ✓ Electric Heater: 1,000 - 3,000 watts
  - ✓ Ceiling Fan: 50 - 100 watts
- Kitchen Appliances:
  - ✓ Refrigerator: 100 - 800 watts
  - ✓ Microwave Oven: 600 - 1,200 watts
  - ✓ Electric Oven: 2,000 - 5,000 watts
  - ✓ Dishwasher: 1,200 - 1,500 watts
  - ✓ Toaster: 800 - 1,500 watts
  - ✓ Blender: 300 - 1,000 watts
  - ✓ Slow Cooker: 200 - 500 watts
  - ✓ Coffee Maker: 600 - 1,200 watts

Electrical Ratings of Air Conditioners

❖Power Consumption

✓1.5 Ton AC: About 1,500 to 2,000 watts (1.5 to 2 kW).

✓2 Ton AC: About 2,000 to 2,500 watts (2 to 2.5 kW).

❖Voltage Measurement:

✓Standard household voltage: 230V in many countries (e.g., Europe, Asia) and 120V in the U.S. for smaller units.

❖Current (Amperage)

✓Current (A) = Power (W) / Voltage (V).

✓1.5 Ton AC: Typically around 7 - 10 A.

✓2 Ton AC: Typically around 10 – 12.

❖Power Factor

✓Ranges from 0 to 1; values closer to 1 indicate better efficiency.

Electrical Ratings of PCs

❖Power Consumption

- ✓Basic PCs: 300 - 500 watts.
- ✓Mid-range gaming PCs: 500 - 800 watts.
- ✓High-end gaming or workstation PCs: 800 - 1,200 watts or more.

❖Voltage Measurement:

- ✓Standard Input Voltage: Typically 100-240V AC, depending on the region.
- ✓Output Voltages: The PSU converts AC to DC voltages used by the PC:
- ✓+12V (for CPU, GPU, and other components)
- ✓+5V (for USB devices and logic circuits)
- ✓+3.3V (for motherboards and some peripherals)

❖Current (Amperage)

- ✓1 to 20 A or more

Electrical Ratings of Laptops

❖Power Consumption

✓Standard laptops: 45 - 65 watts.

✓Gaming laptops: 120 - 230 watts.

❖Voltage Measurement:

✓Input Voltage: Most laptop adapters operate within the range of 100V to 240V AC for international compatibility.

✓Output Voltage: Typically ranges from 15V to 20V DC, depending on the laptop model.

❖Current (Amperage)

✓Current (A) = Power (W) / Voltage (V).

✓Standard laptops may draw 2-4 A during normal operation.

✓Gaming laptops may draw 6-12 A or more during intensive tasks.

Electrical Ratings of Printers

- Power Consumption
  - ✓Inkjet Printers: Typically consume 30 - 100 watts during printing.
  - ✓Laser Printers: Generally consume 300 - 800 watts during printing.
- Voltage Measurement:
  - ✓100-240V AC for international compatibility.
- Current (Amperage)
  - ✓Current (A) = Power (W) / Voltage (V).
  - ✓Smaller inkjet printers may draw about 0.5 - 1.5 A.
  - ✓Larger laser printers may draw 1 - 4 A, especially during peak usage.

Types of Tariff

❖Today's interconnected power systems supply a number of consumers. With such a big organization, management, economy and control come into account automatically. The supply companies (usually in the public sector) have to sell their electricity at such a rate that it covers the costs of generation, transmission, distribution, the salaries of the employees, the interest and depreciation and the profit targeted by the company. This rate at which electrical energy is sold to the consumers is termed as "tariff".

❖**Various types of electricity tariff**

1.Simple tariff

2. Flat rate tariff

3.Block rate tariff

4.Two part tariff

5.Maximum demand tariff

6.Power factor tariff

7.Three part tariff

Block rate tariff

❖In this tariff, the first block of the energy consumed (consisting of a fixed number of units) is charged at a given rate and the succeeding blocks of energy (each with a predetermined number of units) are charged at progressively reduced rates. The rate per unit in each block is fixed.

❖For example, the first 50 units (1st block) may be charged at 3 rupees per unit; the next 30 units (2nd block) at 2.50 rupees per unit and the next 30 units (3rd block) at 2 rupees per unit.

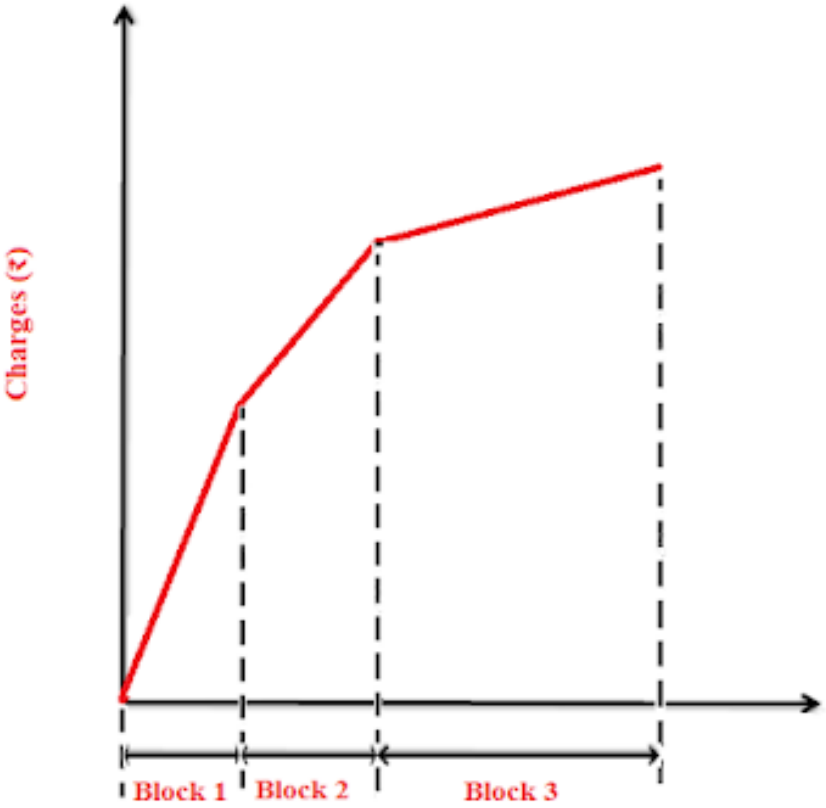


Figure : Block Rate Tariff

ENERGY CONSUMPTION CALCULATION

❖Energy and power are closely related. Electrical energy can be measured only when electrical power is known. So, first, we understand the electrical power. Electrical power is the amount of electrical current that results from a certain amount of voltage or we can say that power is the rate at which energy is delivered. It is measured in watts.

❖Mathematically it is written as Power =Voltage x Current

❖ The measurement of electrical energy is completely dependent on power which is measured in watt, kilowatts, megawatts, gigawatts, and time which is measured in an hour. Joule is the smallest unit of energy. But for some bigger calculation, some better unit is required. So, the unit used for electrical energy is watt-hour.

❖Electrical energy is the product of electrical power and time, and it is measured in joules. It is defined as “1 joule of energy is equal to 1 watt of power is consumed for I second". i.e.Energy= Power × Time ;1 Joule =1 Watt x 1 sec .

❖Watts are the basic unit of power in which electrical power is measured or we can say that rate at which electric current is being used at a particular moment.

ENERGY CONSUMPTION CALCULATION

❖Watt-hour is the standard used for measurement of energy, describing the amount of watts used over time. It shows how fast the power is consumed in the period of time.

❖Energy in Watt hours = Power in Watts x Time in hours .

❖Kilowatt-hour is simply a bigger unit of energy when large appliances drawn power in kilowatts. It can be described as one kilowatt hour is the amount of energy drawn by the 1000 watts appliance when used for an hour.

❖Where, One kilowatt =1000 watts

❖Energy in kiloWatt hours = Power in KiloWatts x Time in hours .

❖The electrical supply companies take electric energy charges from their consumer per kilowatt hour unit basis.

Calculation of electricity bill for domestic consumers

❖The calculation of an electricity bill for domestic consumers typically involves a straightforward process based on the following components:

(i) Consumption: The primary factor is the amount of electricity consumed, measured in kilowatt-hours (kWh). This is determined by the electricity meter installed at your premises.

(ii) Tariff rate: The cost per unit of electricity (kWh) is specified by your electricity provider and may vary depending on your location and the tariff plan you are on.There might be different rates for different usage tiers or times of day.

Calculation of electricity bill for domestic consumers

Example: A consumer uses a 10kW geezer, a 6kW electric furnace and five 100W bulbs for 15 hours. How many units (kWh) of electrical energy have been used and what would be the electricity bill?

Explanation: Given that

Load-1=10 kW geezer

Load-2=6 kW electric furnace

Load-3=500 watt (five 100 watt bulbs)

Total load= 10 kW +6 kW + 0.5 kW

Time taken =15 hours

Energy Consumed= Power in kW x Time in hours = 16.5 x 15 =247.5 units

Calculation of electricity bill for domestic consumers

Example: A consumer uses a 10kW geezer, a 6kW electric furnace and five 100W bulbs for 15 hours. How many units (kWh) of electrical energy have been used and what would be the electricity bill?

For above electrical energy consumption, the tariff can be calculated as follows:

1 unit = 1kWh

So, the total energy consumption = 247.5 units

If the cost per unit is Rs.2.5, then the total electricity bill= 247.5x2.5= Rs.618.75/-

Calculation of Electricity Bill for Domestic Consumers

❖Let us consider different home appliances to calculate approximate total energy consumption of house per month.

S. No.	Appliances	Watts	No	Total no watts	No. of operational hours per day	Total energy consumed = No of Watts × No. of operation hours	Energy consumed in kWh (units) per day= energy consumed /1000
1.	Tube light	60	10	600	5	3000	3
2.	Fan	75	4	300	8	2400	2.4
3.	Refrigerator	200	1	200	24	4800	4.8
4.	AC	1000	1	1000	5	5000	5
5.	Laptop	50	1	50	2	100	0.1
6.	Television	50	1	50	3	150	0.15
7.	Grinders	1000	1	1000	½	500	0.5
8.	Printers	50	1	50	½	25	0.025
9.	Washing Machine	2000	1	2000	1	2000	2
10.	Micro wave	1000	1	1000	1	1000	1
						Total number of units consumed per day	18.9 ≈ 19 units

FUSE

The electrical equipment's are designed to carry a particular rated value of current under normal conditions. Under abnormal conditions such as short circuit, overload or any fault; the current raises above this value, damaging the equipment and sometimes resulting in fire hazards. Fuses are come into operation under fault conditions.

A fuse is a short piece of metal, inserted in the circuit, which melts when excessive current flows through it and thus breaks the circuit. Under normal operating conditions it designed to carry the full load current. If the current increases beyond this designed value due to any of the reasons mentioned above, the fuse melts and thus isolating the power supply from the load.

FUSE

(a) Desirable characteristics of a fuse element: The material used for fuse wires must have the following characteristics:

Low melting point e.g., tin, lead.

High conductivity e.g., silver, copper.

Free from deterioration due to oxidation e.g., silver.

Low cost e.g., lead, tin, copper.

(b) Materials: Materials used are tin, lead or silver having low melting points. Use of copper or iron is dangerous, though tinned copper may be used.

(c) Types of fuses: Fuses are classified into following types:

(i) Re-wireable or kit-kat fuse and

(ii) High rupturing capacity (HRC) cartridge fuse

Re-wireable or kit-kat fuse

Re-wireable fuse is used where low values of fault current are to be interrupted. These fuses are simple in construction, cheap and available up to a current rating of 200A. They are erratic in operation and their performance deteriorates with time. An image of re-wireable fuse is as shown in figure .

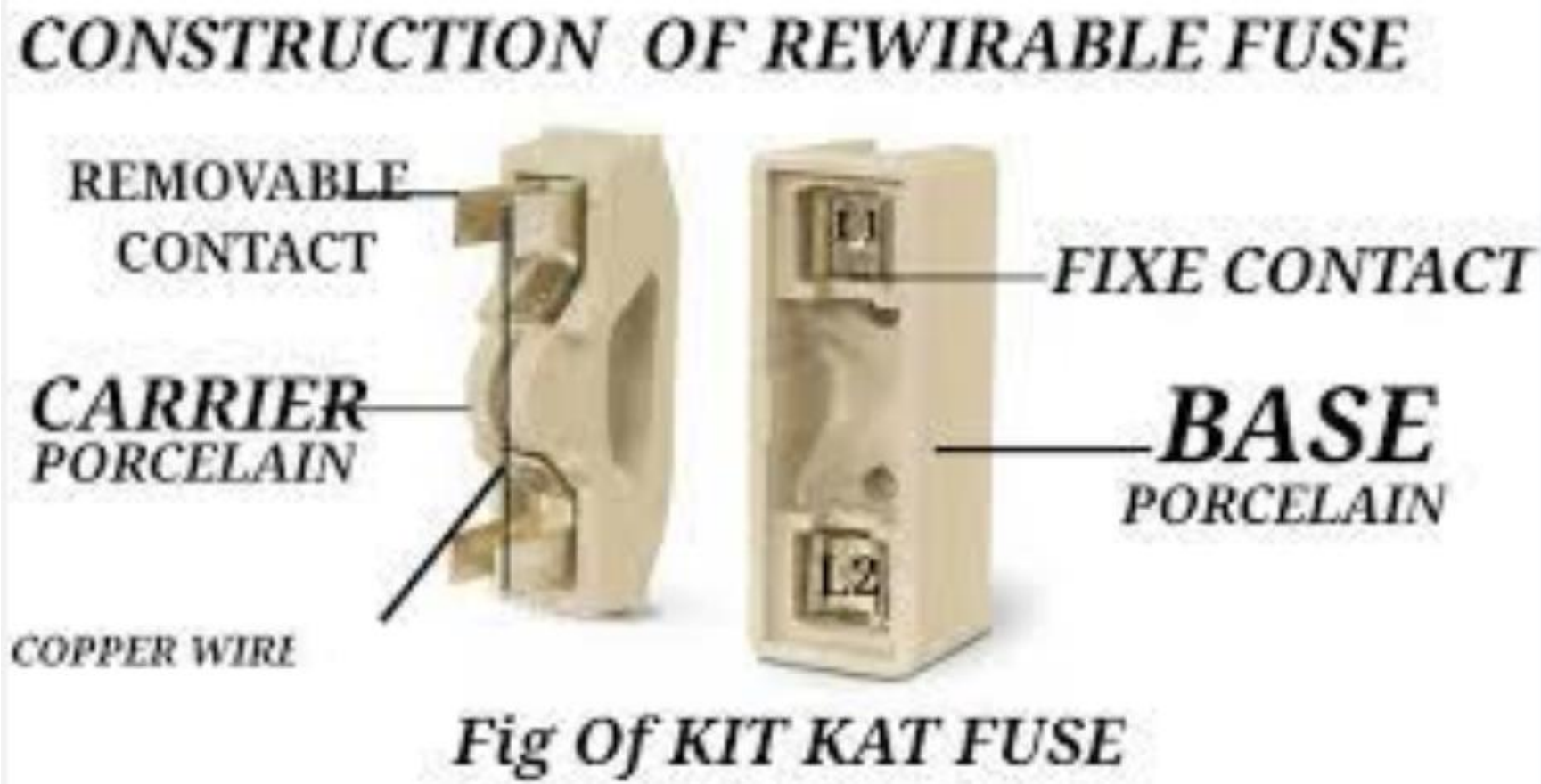


Fig Of KIT KAT FUSE

High Rupturing Capacity (HRC) Cartridge Fuse

Figure shows an image of HRC cartridge fuse and figure ,shows the essential parts of a typical HRC cartridge fuse. It consists of a heat resisting ceramic body having metal end-caps to which a silver current-carrying element is welded. The space within the body surrounding the element is completely packed with a filling powder. The filling material may be chalk, plaster of paris, quartz or marble dust and acts as an arc quenching and cooling medium. Therefore, it carries the normal current without overheating.

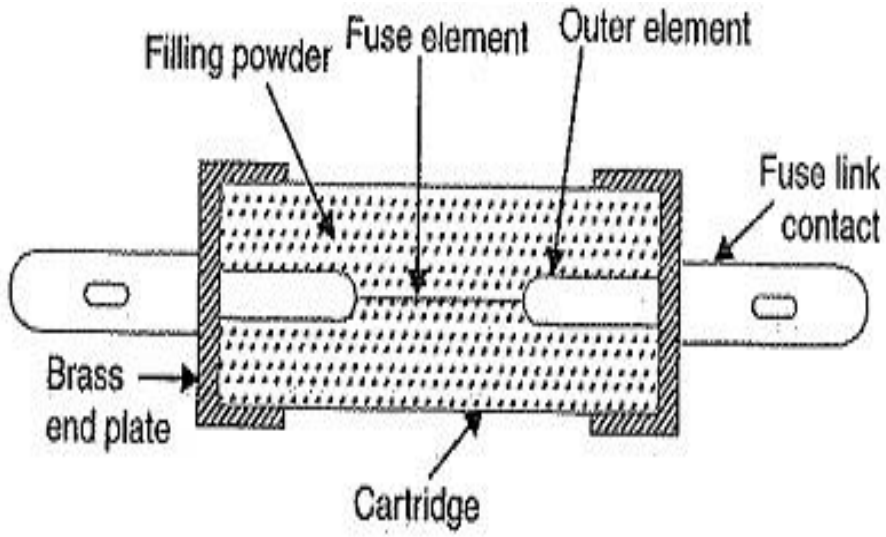


Fig. 20.3

High Rupturing Capacity (HRC) Cartridge Fuse

Under normal loading conditions, the fuse element is at a temperature below its melting point. When a fault occurs, the current increases and the fuse element melts before the fault current reaches its first peak. The heat produced in the process vaporizes the melted silver element. The chemical reaction between the silver vapour and the filling powder results in the formation of a high resistance substance which helps in quenching the arc.

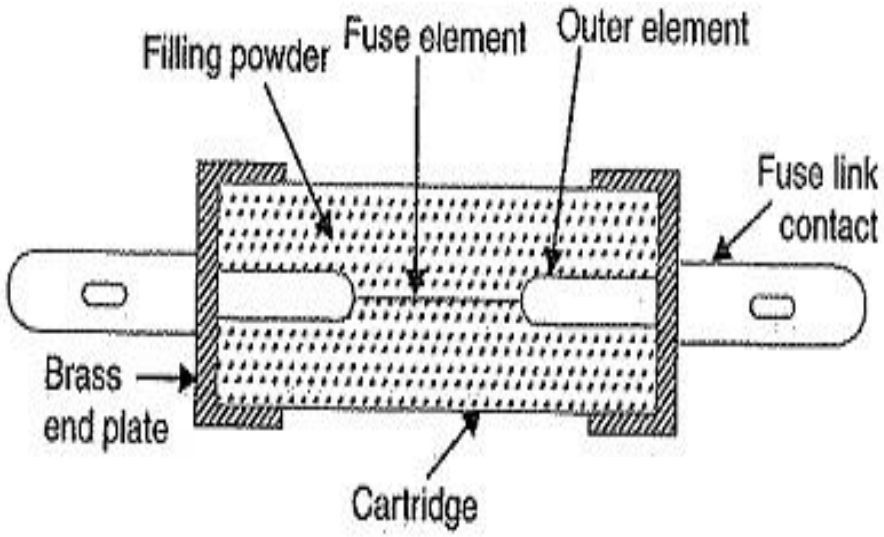


Fig. 20.3

Circuit Breaker

Circuit Breaker Definition: A circuit breaker is a manually or automatically operated electrical switch designed to protect and control power systems by interrupting fault currents.

How Circuit Breakers Work: By detecting faults like overloads or short circuits, circuit breakers interrupt the current flow, activate arc quenching methods, and can be reset to reinstate the flow of electricity.

Types of Circuit Breakers: The choice of a circuit breaker type—MCB,oil, air, SF6, vacuum—depends on the application's voltage level, current rating, and environmental conditions.

Resetability and Reuse: Unlike fuses, circuit breakers can be reset and reused, making them more sustainable and cost-effective.

Circuit Breaker Safety: Circuit breakers enhance electrical system safety by preventing equipment damage, fire hazards, and power outages.

Miniature circuit breaker (MCB).

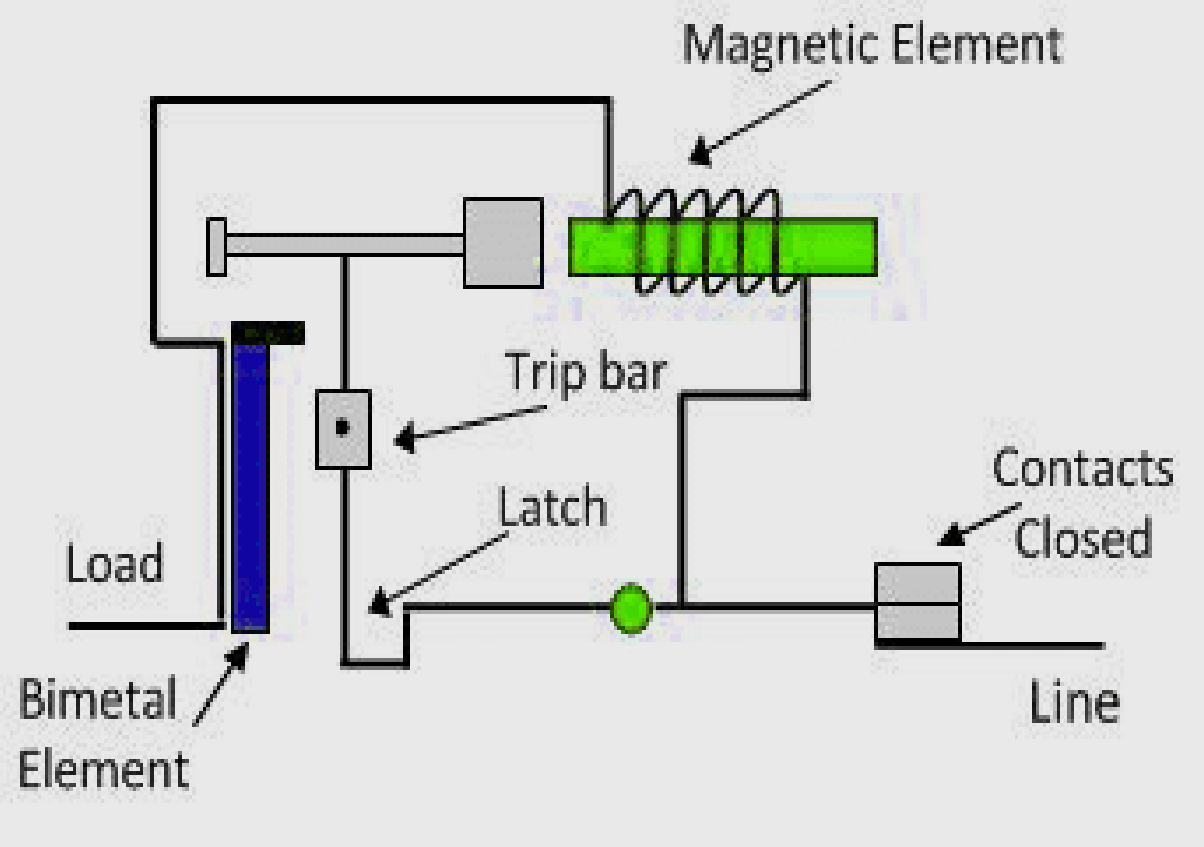
- A miniature circuit breaker (MCB) is an Electrical Switch that automatically switches off the electrical circuit during an abnormal condition of the network means an overload condition as well as a faulty condition.
- Nowadays we use an MCB in a low-voltage electrical network instead of a fuse. The fuse may not sense it but the miniature circuit breaker does it in a more reliable way. MCB is much more sensitive to overcurrent than a fuse.
- Handling an MCB is electrically safer than a fuse. Quick restoration of supply is possible in case of a fuse because fuses must be rewirable or replaced for restoring the supply. Restoration is easily possible by just switching it ON. Let's look at the working of the miniature circuit breaker.

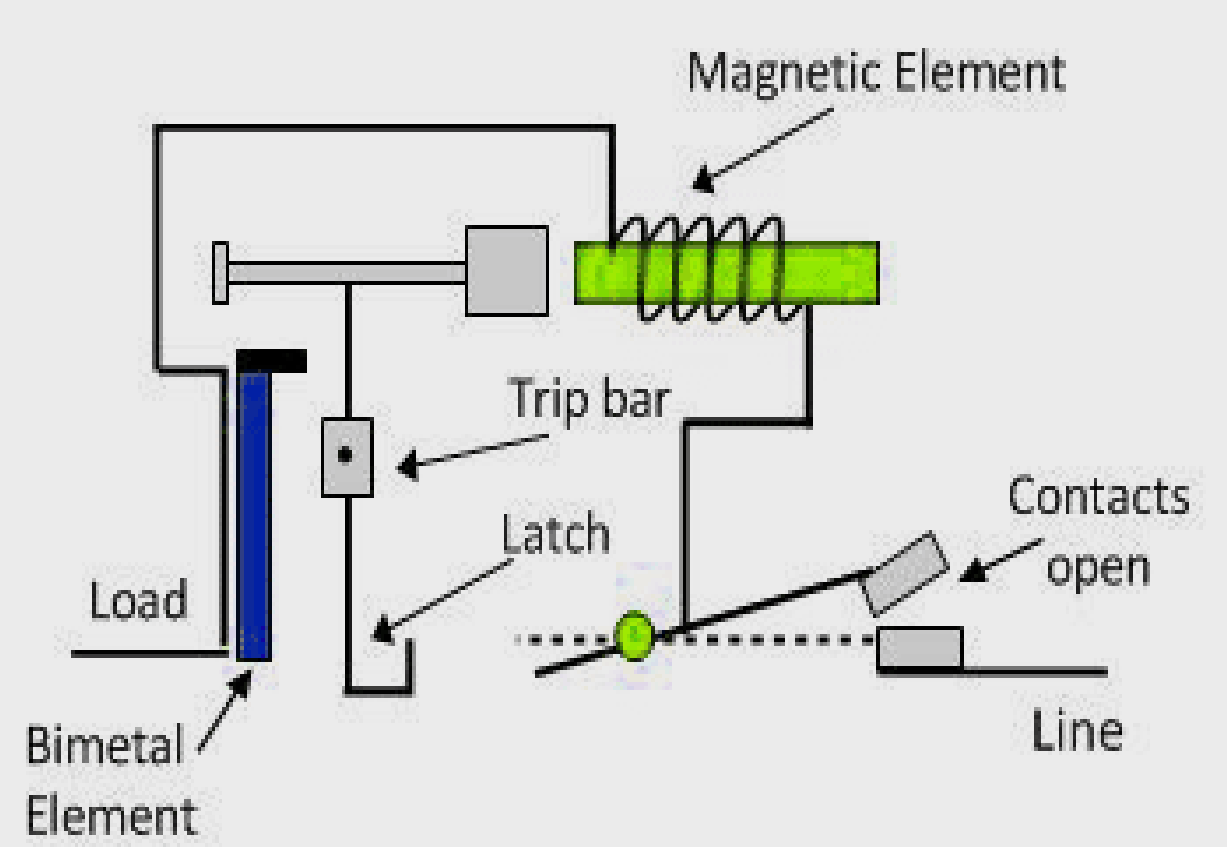


Miniature circuit breaker (MCB)-Working

- Whenever continuous overcurrent flows through MCB, the bimetallic strip is heated and deflects by bending. This deflection of the bi-metallic strip releases a mechanical latch.
- As this mechanical latch is attached to the operating mechanism, it causes to open the miniature circuit breaker contacts, and the MCB turns off thereby stopping the current to flow in the circuit. To restart the flow of current the MCB must be manually turned ON.
- An MCB is very simple, easy to use, and is not generally repaired. It is just easier to replace. The trip unit is the main part, responsible for its proper working. There are two main types of trip mechanisms.
- A bi-metal provides protection against overload current and an electromagnet provides protection against electric short-circuit current.

Miniature circuit breaker (MCB)-Working





Miniature circuit breaker (MCB)-Operation

- If the circuit is overloaded for a long time, the bi-metallic strip becomes overheated and deformed. This deformation of the Bi-metallic strip causes displacement of the latch point.
- The moving contact of the MCB is arranged by means of spring pressure, with this latch point, a little displacement of the latch causes, the release of spring and makes the moving contact move for opening the MCB.
- The current coil or trip coil is placed so that during a short circuit fault the magneto-motive force (mmf) of the coil causes its plunger to hit the same latch point and make the latch to be displaced.
- Again, when the operating lever of the miniature circuit breaker is operated by hand, that means when MCB goes off position manually, the same latch point is displaced as a result of moving contact separated from fixed contact in the same manner.

Miniature circuit breaker (MCB)-Operation

- It may be due to the deformation of a bi-metallic strip, increased mmf of a trip coil, or maybe a manual operation, the same latch point is displaced and the same deformed spring is released, which is ultimately responsible for the movement of the moving contact. When the moving contact is separated from fixed contact, there may be a high chance of arc.
- This arc then goes up through the arc runner and enters arc splitters and is finally quenched. When we switch it on, we reset the displaced operating latch to its previous on position and the MCB is ready for another switch off or trip operation.

Advantages of Circuit Breaker.

- Be reset and reused multiple times without replacement
- Be operated manually or automatically for switching purposes
- Provide selective protection and isolation of different parts of the power system
- Interrupt high fault currents without producing harmful gases or flames
- Reduce the risk of fire hazards and power outages caused by faults

Disadvantages of Circuit Breaker

- They are more expensive and complex than fuses
- They require regular maintenance and testing to ensure the proper functioning
- They may fail to operate due to mechanical or electrical faults
- They may cause voltage surges or transients during switching operations

Electric shock

- Electric shock occurs when an electric current passes through the body, potentially causing injury or even death. The severity of an electric shock depends on several factors, including the voltage, current, duration of exposure, and the path the current takes through the body.
- **Causes of Electric Shock**

✓ Direct Contact: Touching live wires or faulty electrical appliances.

✓ Faulty Equipment: Defective wiring, damaged cords, or malfunctioning devices.

✓ Wet Conditions: Water is a good conductor; using electrical devices in wet environments increases the risk.

✓ Improper Use: Using electrical devices without following safety guidelines or using them in unintended ways.

Electric shock

- The severity of the injuries that ensue include:
- ✓ The voltage involved.
- ✓ The amperage of the electric current involved.
- ✓ The pathway that the electric charge took in the person's body (e.g., through the heart, muscles, head, eyes and/or chest, or hand-to-hand).
- ✓ How long the victim had contact with the electrical source.
- ✓ What the health and/or medical condition of the victim was prior to the shock.
- ✓ Whether the electrical current was direct (DC) or alternating (AC).

Preventive Methods to Avoid Electric Shock

- Preventing electric shock is crucial for ensuring personal safety, whether at home or in a workplace. Here are some preventive methods to avoid electric shock:

(i) Regular maintenance

(ii) Use Ground Fault Circuit Interrupters (GFCIs)

(iii) Follow the manufacturer's instructions

(iv) Shut-off the power supply

(v) Test for live wires

(vi) Use insulated tools while working

(vii) Follow safety guidelines

(viii) Outlet Covers

(ix) Inspect cords and cables

(x) Keep a safe distance

Preventive Methods to Avoid Electric Shock

(xi) Proper outlet usage

(xii) Extension cords

(xiii) Stay dry

(xiv) Educate yourself

(xv) Use grounding

(xvi) Display danger board

(xvii) Avoid overhead power lines

(xviii) Emergency planning

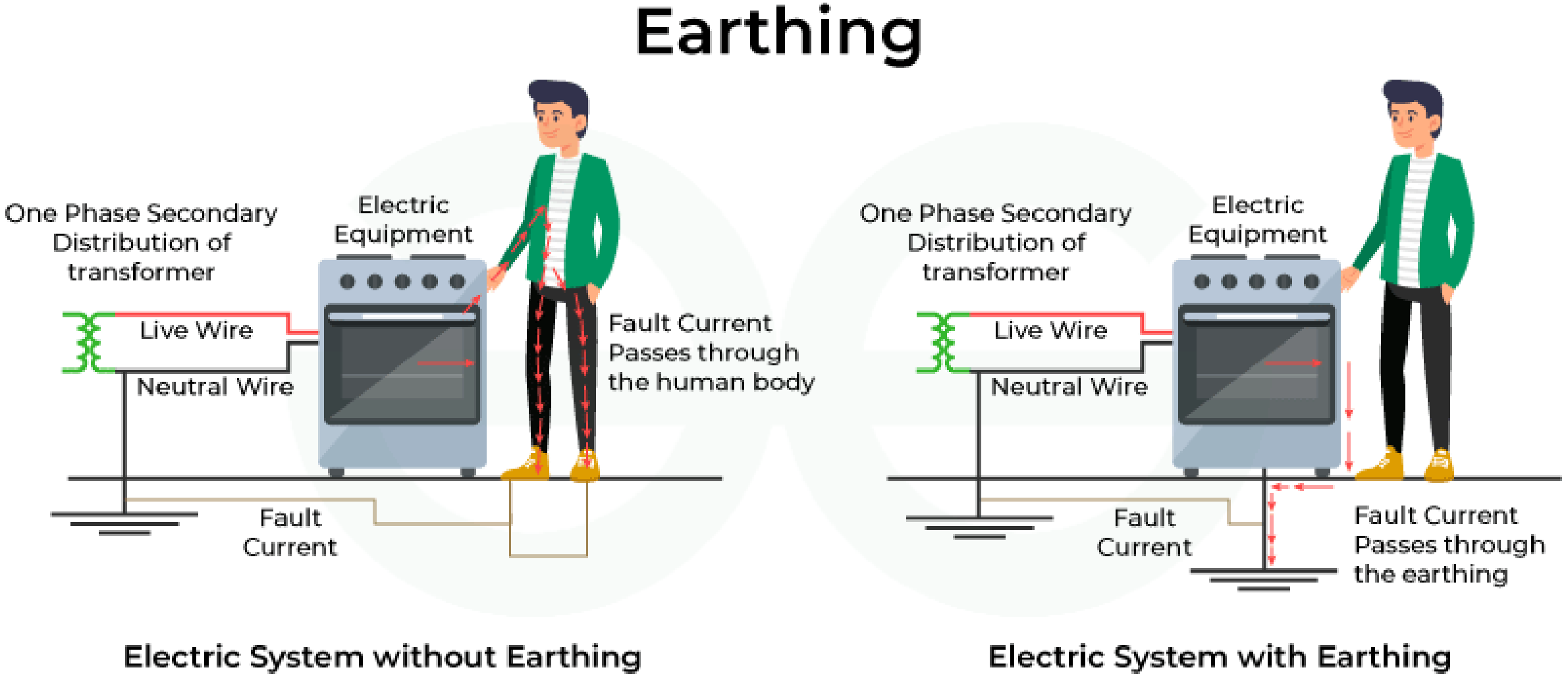
Safety Precautions to avoid shock

- Avoid water at all times when working with electricity. Never touch or try repairing any electrical equipment or circuits with wet hands. It increases the conductivity of the electric current.
- Never use equipment with damaged insulation. The insulation of conductors must be proper and in good condition.
- Earth connection should be maintained in proper condition
- Use of the fuses and cables of proper rating.
- Use the rubber soled shoes while working.
- Megger tests should be done to check the insulation.
- Never touch two different terminals at the same time.
- Never remove the plug by pulling wire.
- The sockets should be placed at a proper height
- Switch off supply and remove the fuses before starting the work with any installation.
- Always use insulated screw drivers, and line testers.

Concept of Earthing or Grounding

- Earthing is a traditional and efficient way of transferring and immediately discharging of electrical energy to the earth directly through a low-resistance wire.
- We have seen that a normal plug consists of 3 pins. These pins are live, neutral, and earth pins. This earth pin is the main pin used for transferring extra charge from devices and thereby protecting the device.
- The pin transfers extra charges to earth through low-resistance wire avoiding any major shock. Thereby Earthing is a major process used in electrical engineering. Earthing is a process in which the leakage current is made to flow to the earth to prevent the accumulation of excess current in a device. Thereby it prevents any major shock or fire due to static charges.
- In scientific terms, we can define earthing as : The process in which the sudden discharge of the electric current takes place by transferring charges directly to the earth's ground using a low resistance wire.

Concept of Earthing



Need for Earthing or Grounding

- Earthing is used for security and overall functionality in electric components. In case of situations like lightning, major voltage surge the current flowing in a device exceeds the optimal value. This can lead to disastrous situations like fire due to static charges, huge electric shocks causing damage of the device. To avoid any such serious mishap , we use earthing.
- The excess current called fault current is made to flow into earth ground which acts as a neutral plate. This ensures there is no deposition of excess current on devices and hence earthing protects devices from getting damaged. Hence, earthing serves several important purposes by contributing to the safety, reliability, and proper functioning of grounding systems.

Types of Earthing or Methods of Earthing

(i) Plate earthing

(ii) Pipe earthing

(iii) Rod earthing

(iv) Strip or wire earthing

ADITYA UNIVERSITY

Thank You